

Business Process Automation with VBA and Python

Mr. Eddie Chow / 6 June 2026



Table Of Contents

Introduction to business process automation

Business Process automation with VBA

Business process automation with Python

Introduction to project management for business process automation



Development and implementation of business process automation

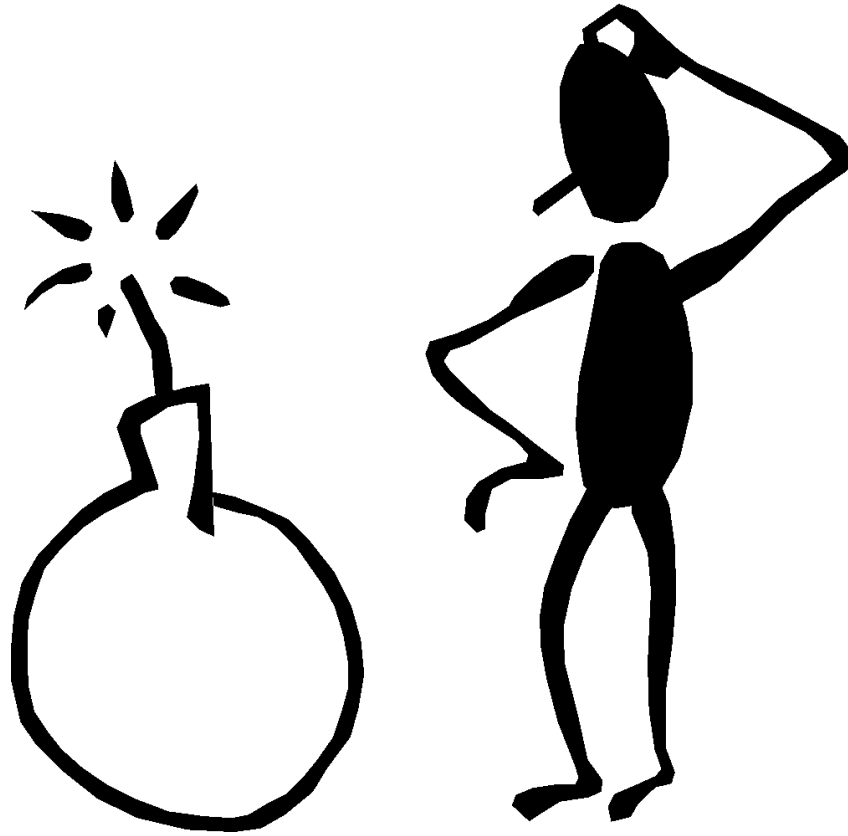
Final Group Presentation





What is Process Automation, RPA, IPA?
Overview of the technological building blocks related to process automation
Contemporary tools for process automation
Challenges and opportunities of business process automation
Business implications of process automation

Project Risk Management



Project Risk – Definition

“Project risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on a project objective”

“A combination of the probability of a defined **threat** or **opportunity (Likelihood)** and the magnitude of the consequences of the occurrence (Impact) defines a Risk Index”

Tasks	Threat	Reason
Data Collection	Delay 2 days	Business Scope Always changing
Data Cleaning	Delay 2 days	Business Scope Always changing
Machine Learning Model	Delay 0 days	Nil

Risk Impact

Threat → Scope → **Poor Quality Product**

Threat → Schedule → **Late Delivery**

Threat → Cost → **Overspend**

In addition there are health, safety and environmental threats that **must** be managed

Identify Risks

Assess likelihood and impact

Rank risks and prioritize

Define risk management approach & actions

Implement actions

Monitor & review

Make the management of risk integral to the way the project is managed

Ensure that cost and time contingencies are consistent with identified risks

Focus on the “significant few” – don’t try to manage too many risks

Be vigilant (be ready for possible problems and difficulties) and proactive

Project Monitoring and Control



Project Monitoring

Typical Monitoring Activities

- regular reviews of progress against schedule using WBS as basis (Plan against Baseline)
- regular review of actual costs (O/P from SAP) against budgeted costs and Earned Value at WBS level
- regular review of resource loading
- regular progress meetings with project team
- regular meetings with contractors
- production of periodic progress reports
- risk reviews
- inspections/ audits



Project Control

Typical Control Activities

- assign responsibilities at Work Package level
- staged authorisation of work to be done
- staged release of budgets (staged release of WBS(e) numbers)
- ensure PM has a 'Management Reserve' under his control
- seek corrective action reports when WPs go 'off track' (overrunning or overspending)
- release Management Reserve carefully



Project Monitoring and Control Summary

Monitor against the plan – status regularly

Take a factual approach to decisions

Identify management action early

Check that defined controls are being applied – correct if necessary

Apply change control

Automation on Stakeholder Engagement process

1. Identify Stakeholders

- Mapping Stakeholders: Begin by identifying all relevant stakeholders, both internal (employees, management) and external (customers, partners, investors). Create a comprehensive stakeholder map to visualize their roles and influence.
- Prioritization: Classify stakeholders based on their level of interest and influence. This will help prioritize engagement efforts and tailor strategies accordingly

2. Assess Needs

- Conduct Surveys and Interviews: Gather information on stakeholders' needs, expectations, and concerns through surveys or one-on-one interviews. This qualitative data is essential for understanding what stakeholders value most.
- Analyze Data: Use analytics tools to interpret the data collected. Identify common themes and specific requirements that can guide your engagement strategy.

3. Define Actions

- Develop Engagement Strategies: Based on the assessed needs, define specific actions tailored to each stakeholder group. This could include personalized communication plans, involvement in decision-making processes, or targeted information sessions.
- Set Clear Goals: Establish measurable objectives for each action to ensure accountability and track progress over time.

Automation on Stakeholder Engagement process

4. Establish Communication Channels

- Choose Appropriate Platforms: Select the most effective communication channels for engaging with different stakeholders. Options may include email newsletters, social media updates, webinars, or dedicated project management tools
- Automate Communication: Utilize automation tools to streamline communication efforts. For example, set up automated email campaigns or notifications to keep stakeholders informed without manual effort

5. Gather Feedback

- Implement Feedback Mechanisms: Create structured methods for collecting feedback from stakeholders after key interactions or milestones. This could involve follow-up surveys or feedback forms.
- Encourage Open Dialogue: Foster an environment where stakeholders feel comfortable sharing their thoughts and concerns. Use tools like online forums or chatbots to facilitate ongoing conversations.

6. Monitor and Review

- Track Engagement Metrics: Regularly monitor key performance indicators (KPIs) related to stakeholder engagement, such as response rates, satisfaction levels, and participation in events.
- Evaluate and Adjust Strategies: Conduct periodic reviews of your engagement strategies based on the feedback received and the metrics tracked. Make necessary adjustments to improve effectiveness and ensure alignment with stakeholder needs

Methodologies in Project Management

1. Waterfall Project Management
 - Waterfall project management follows a linear and sequential process where each phase must be completed before the next begins. It suits projects with fixed requirements and clear objectives.
2. Agile Project Management
 - Agile project management is an iterative approach that emphasizes flexibility, customer collaboration, and rapid delivery through short cycles called sprints, allowing teams to adapt quickly to changing requirements
3. Hybrid Project Management
 - Hybrid project management combines Agile and Waterfall methodologies, allowing teams to use structured planning while maintaining flexibility for iterative development, making it ideal for projects needing both predictability and adaptability.
4. Other Notable Methodologies
 - Other methodologies include Scrum, an Agile framework focusing on short cycles and team roles, and Lean, which maximizes value by minimizing waste and improving processes throughout the project lifecycle.

Successful Automation Projects: Case Studies

- **UiPath Automation Case Studies**

1. Enerjisa (Energy): Improved internal processes and customer service, leading to enhanced operational efficiency and customer satisfaction.
2. Suncoast Credit Union (Financial Services): Utilized AI and automation to improve member trust and accelerate growth, streamlining banking processes.
3. MongoDB (Technology): Saved over 150,000 working hours and \$1.5 million by automating time-consuming data management tasks.

- **Kawasaki Robotics Case Studies**

1. German Brewery (Food & Beverage): Implemented robotic palletizing systems to reduce labor costs and increase production capacity.
2. Automated Bottle Production Line (Food & Beverage): Enhanced material handling and palletizing processes, resulting in streamlined operations and reduced production times.

- **Ittransition RPA Use Cases**

1. Heritage Bank (Banking): Automated 80 processes across departments, significantly improving operational efficiency and reducing customer interaction processing times.
2. Thermo Fisher Scientific (Biotechnology): Achieved a 70% reduction in invoice processing time for 824,000 invoices annually, enhancing accuracy in financial operations.

Successful Automation Projects: Case Studies

- **Nividous RPA Case Studies**

1. Manufacturer Invoice Processing: Automated data extraction from invoices, saving over \$90,000 annually and speeding up processing times
2. Healthcare Patient Claims Processing: Reduced handling time by 70% through automation of data extraction and claims submission, improving cash flow management.

- **Industrial Automation Insights**

1. JR Automation & Advanced Drainage Systems (Manufacturing): Developed a flexible pipe sorting system that improved operational efficiency while reducing manual labor needs.
2. Snowboard Manufacturing Digital Transformation: Streamlined manufacturing processes using automation technologies, increasing productivity and reducing time-to-market.

Benefits of Using Automation in Project Management

1. Establishing Risks to be Managed

- Real-Time Risk Monitoring: Automation allows for continuous monitoring of project parameters, enabling teams to identify potential risks as they arise rather than after the fact. This proactive approach enhances the ability to mitigate risks effectively before they escalate
- Data Analysis and Predictive Insights: Automated systems can analyze large volumes of data to identify trends and patterns that may indicate emerging risks. This predictive capability allows project managers to prepare for potential challenges and implement strategies accordingly

2. Establishing Costs and Durations

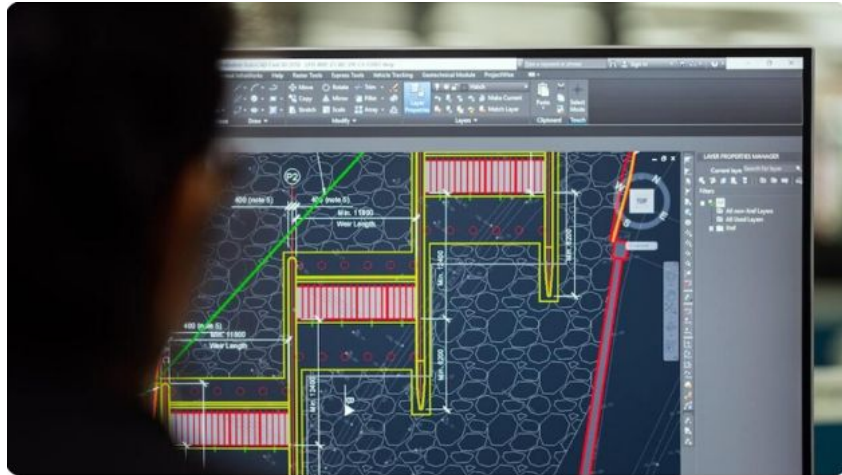
- Accurate Cost Estimation: Automation tools can analyze historical data and current project metrics to provide more accurate cost estimations. This helps in budgeting and financial planning by reducing guesswork and improving reliability
- Streamlined Reporting: Automated reporting features generate cost and duration reports quickly, providing stakeholders with timely insights into project status without manual compilation efforts. This enhances transparency and facilitates informed decision-making

3. Establishing Resources Required

- Resource Optimization: Automation helps in tracking resource allocation and utilization across projects. By analyzing resource usage patterns, organizations can optimize their workforce and materials, ensuring that resources are used efficiently
- Scalability: Automated systems can easily scale to accommodate changes in resource requirements as projects grow or shrink. This adaptability ensures that organizations do not overcommit or underutilize their resources

Defining the Future of Efficiency

- **Definition:** Business process automation (BPA) refers to the use of technology to automate repetitive, manual tasks in business operations to increase efficiency and accuracy.
- **Benefits:** The advantages of BPA include reduced operational costs, increased speed and accuracy of processes, and the ability to reallocate human resources to more strategic tasks.
- **Tools:** Various tools such as workflow automation software, robotic process automation, and AI are instrumental in facilitating BPA by removing bottlenecks in business operations.
- **Examples:** Common applications include automating invoice processing, customer relationship management (CRM), and supply chain management, demonstrating BPA's versatility across industries.



Integration of Project Management and Business Process Automation

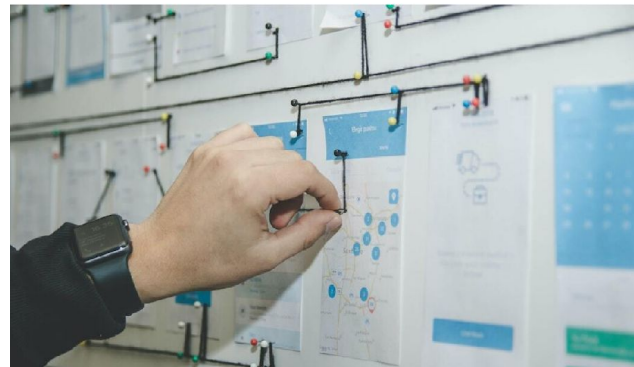
Creating Cohesion for Success

Scope	Description
Synergy	The integration of project management with BPA leads to improved planning, execution, and monitoring of initiatives, enhancing overall project success.
Process Mapping	Visualizing workflows and processes aids in identifying inefficiencies and opportunities for automation, aligning project management initiatives with business objectives.
Best Practices	Employing best practices such as stakeholder engagement, risk management, and iterative development can significantly improve project outcomes in automation.
Alignment	Ensuring that project management frameworks align with automation goals fosters a culture of continuous improvement and operational excellence within the organization.

Integration of Project Management and Business Process Automation

Understanding the Fundamentals

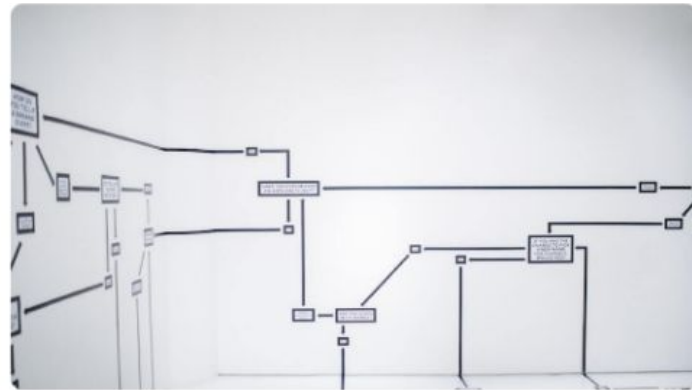
- **Definition:** Project management is the discipline of planning, organizing, and overseeing the successful execution of projects in the context of business process automation, which aims to optimize workflows and improve efficiency.
- **Importance:** The significance of project management in business process automation lies in its ability to streamline processes, enhance productivity, and ensure resources are effectively utilized to meet business objectives.
- **Goals:** The primary objectives encompass enhancing efficiency, reducing costs, improving quality and ensuring successful project delivery that meets stakeholder expectations.
- **Overview:** An effective project management approach integrates all phases of automation projects, from inception through to evaluation, thus providing a comprehensive roadmap for implementation.



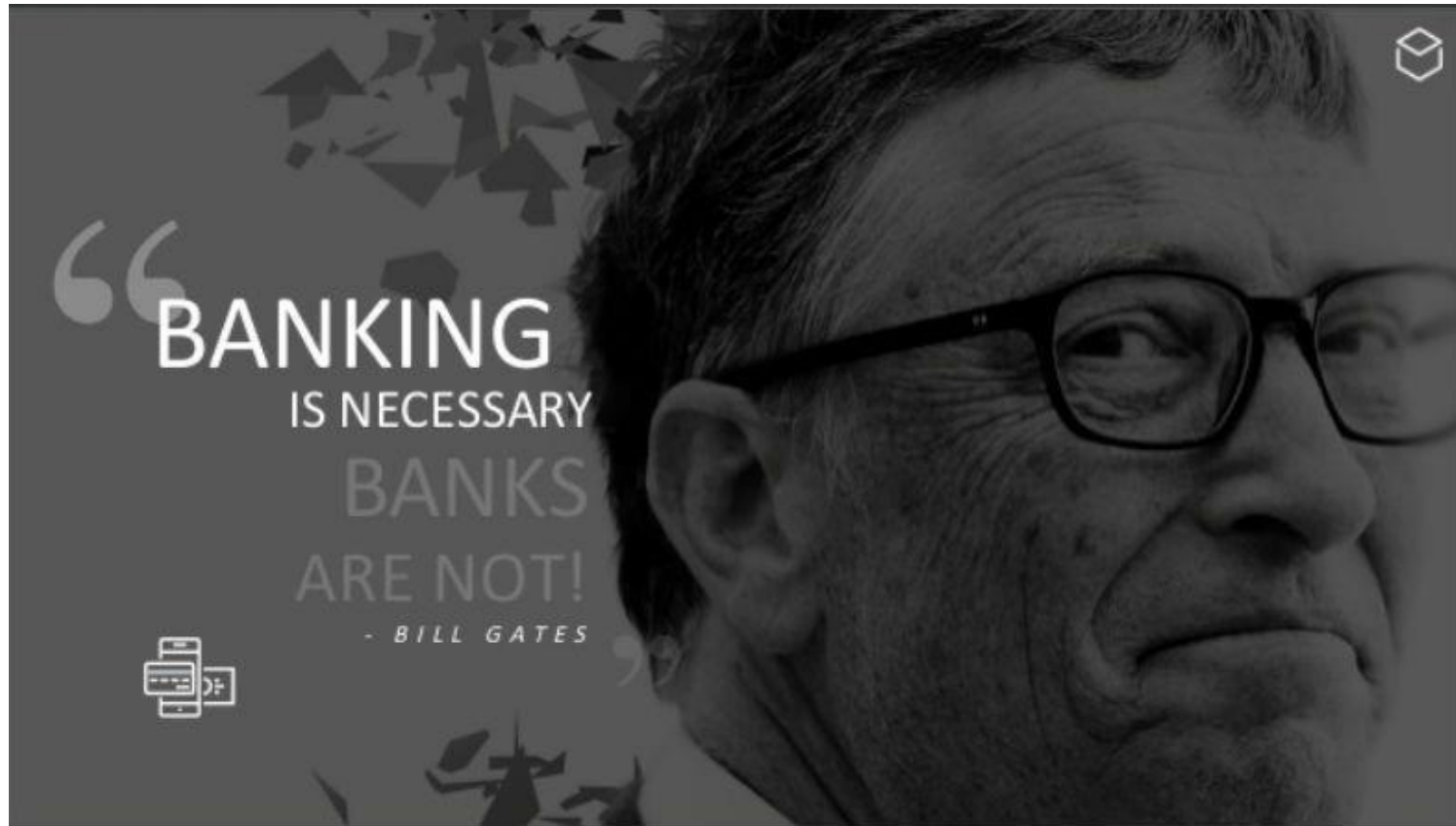
Key Steps in Project Management

The Phases of Project Execution

- Initiation: Defining the project scope, objectives, and stakeholders to establish a shared understanding of the project goals and outcomes.
- Planning: Creating a detailed project plan that outlines tasks, timelines, resources, and budget, ensuring a clear pathway toward project completion.
- Execution: Implementing the project plan, coordinating teams, and managing resources to deliver the project output effectively.
- Monitoring: Continuous oversight of project progress, ensuring alignment with goals, addressing any deviations, and making necessary adjustments to stay on track.
- Closing: Finalizing all project elements, conducting evaluations, and documenting lessons learned to inform future projects and ensure comprehensive closure.



Worldwide Open Banking Movement

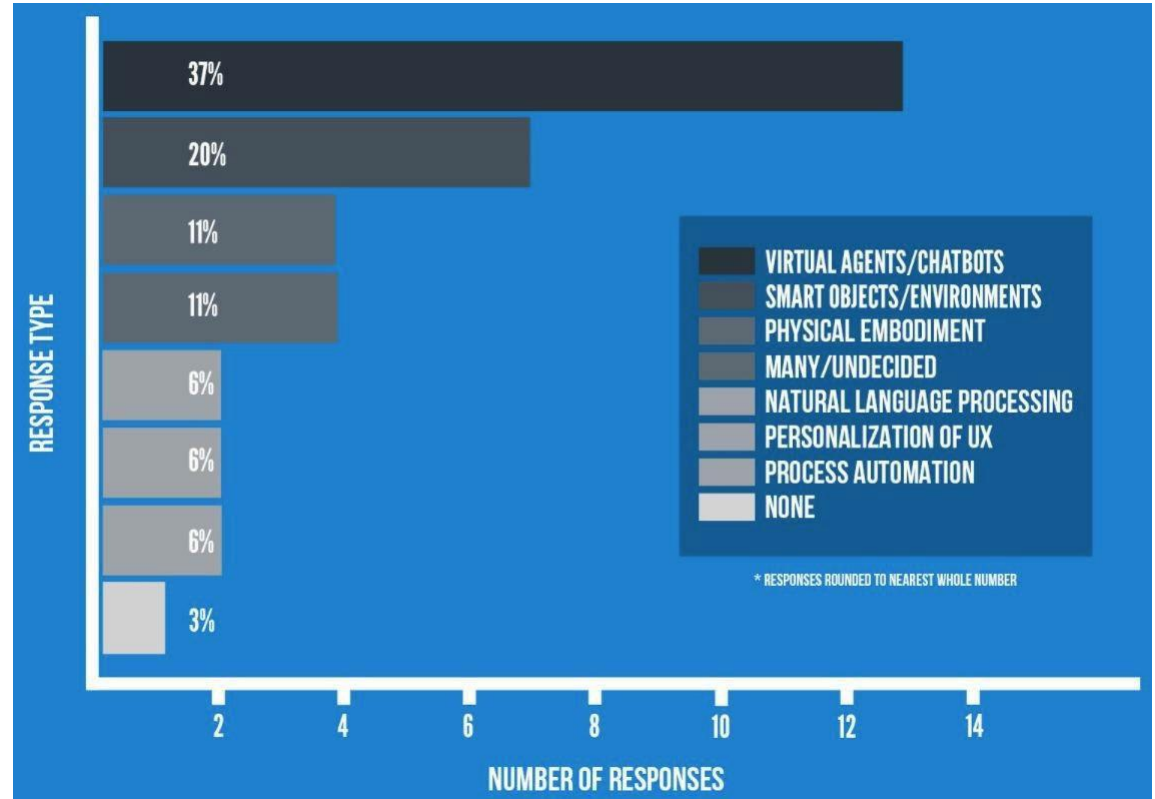


Worldwide Open Banking Movement



<https://www.slideshare.net/CAinc/case-study-open-banking-apis-and-digital-transformationthe-banco-original-story>

Application of AI In Banking: A Case Study



<https://www.techemergence.com/ai-in-banking-analysis/> October 24, 2017 by [Kumba Sennaar](#)

Application of AI In Banking: A Case Study JPMorgan Chase Bank



<https://www.techemergence.co.m/a i -in-banki ng-analysis/> October 24, 2017 by [Kumba Sennaar](#)

Application of AI In Banking: A Case Study



https://twitter.com/_/status/790943639933366274

Chatbot enabled mobile banking has seen steady growth from 12 million customers in 2012 to nearly 22 million in 2016

erica

Based on your typical monthly spending, you have an additional \$150 you could be putting toward your Cash Rewards Visa.

This could save you up to \$300 per year.

Additional Payment

\$150/mn

\$0 ————— \$1540

Payment Details

Current	New	Annual Savings
\$250	\$400	\$300

Next Scheduled Payment: **Nov 17** 📅

Erica - Chatbot is designed to be accessible to clients 24/7 and perform “day-to-day transactions” in addition to **anticipating the unique financial needs of each customer** and helping them reach their financial goals by providing **smart recommendations**

<https://www.cnn.com/2016/10/24/bank-of-america-launches-ai-chatbot-erica-heres-what-it-does.html>

<https://www.bofa.ml.com/en-us/technology-innovation-summit-2017.html>

Application of AI In Banking: A Case Study

Bank of New York Mellon Bank: Software Automation Using Robots

blueprism®

Helped build/deploy 220 software robots to handle repetitive tasks, such as “data requests from external auditors” and “funds transfer bots” which help “correct formatting and data mistakes in requests for dollar funds transfers.



- 100 percent accuracy in account-closure validations across five systems
- 88 percent improvement in processing time
- 66 percent improvement in trade entry turnaround time
- ¼-second robotic reconciliation of a failed trade vs. 5-10 minutes by a human

“funds transfer bots” alone is responsible for \$300,000 in annual savings

<https://www.blueprism.com/news/automation/bny-mellon-became-pioneer-software-robots>

Application of AI In Banking: A Case Study Royal Bank of Scotland: Chatbots

Luvo chatbot pilot is accessible to 50,000 Bank of Scotland iOS mobile customers

answer common queries, such as unknown account transactions, how to make payments and what to do about lost or stolen cards



[https://www.rbs.com/rbs/news/2016/03/rbs-installs-advanced-human-ai-to-help-staff-answer-customer- que.html](https://www.rbs.com/rbs/news/2016/03/rbs-installs-advanced-human-ai-to-help-staff-answer-customer-que.html)

Hottest Application of Predictive Analytics: Vehicle Insurance Fraud Detection



<https://www.youtube.com/watch?v=fELBBWgd6NA>

Watch a youtube video -

CANATICS – Vehicle Insurance Fraud Detection through Data Analytics

Discuss in your group

- What are the limitations of the Canatics Approach?
- How would a smarter criminal defeat Canatics approach?

Hottest Application of Predictive Analytics?



Fraud/Spam Detection



NLP



Speech Recognition

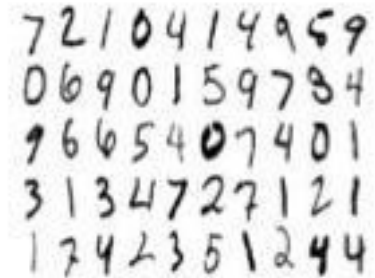
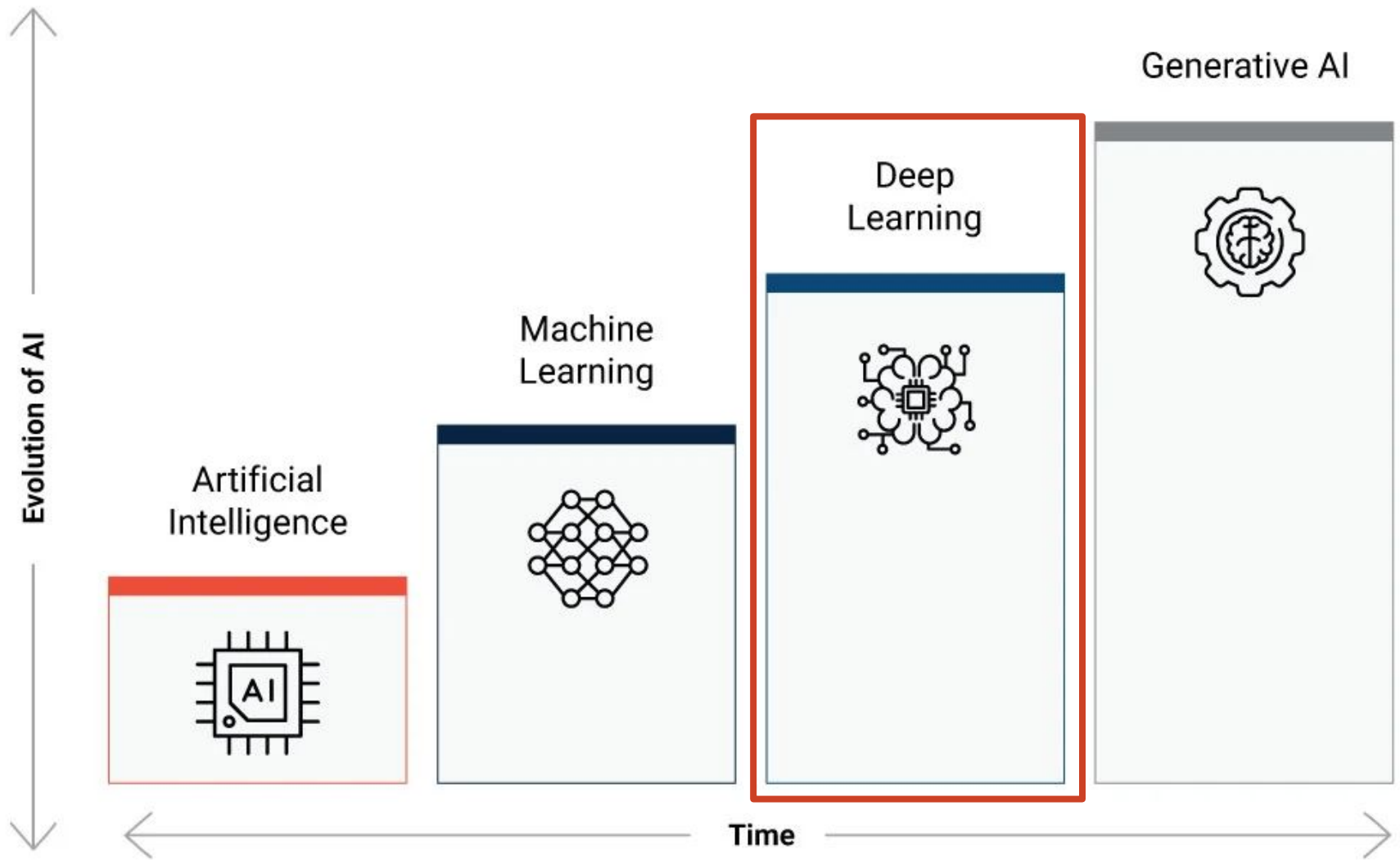


Image Recognition

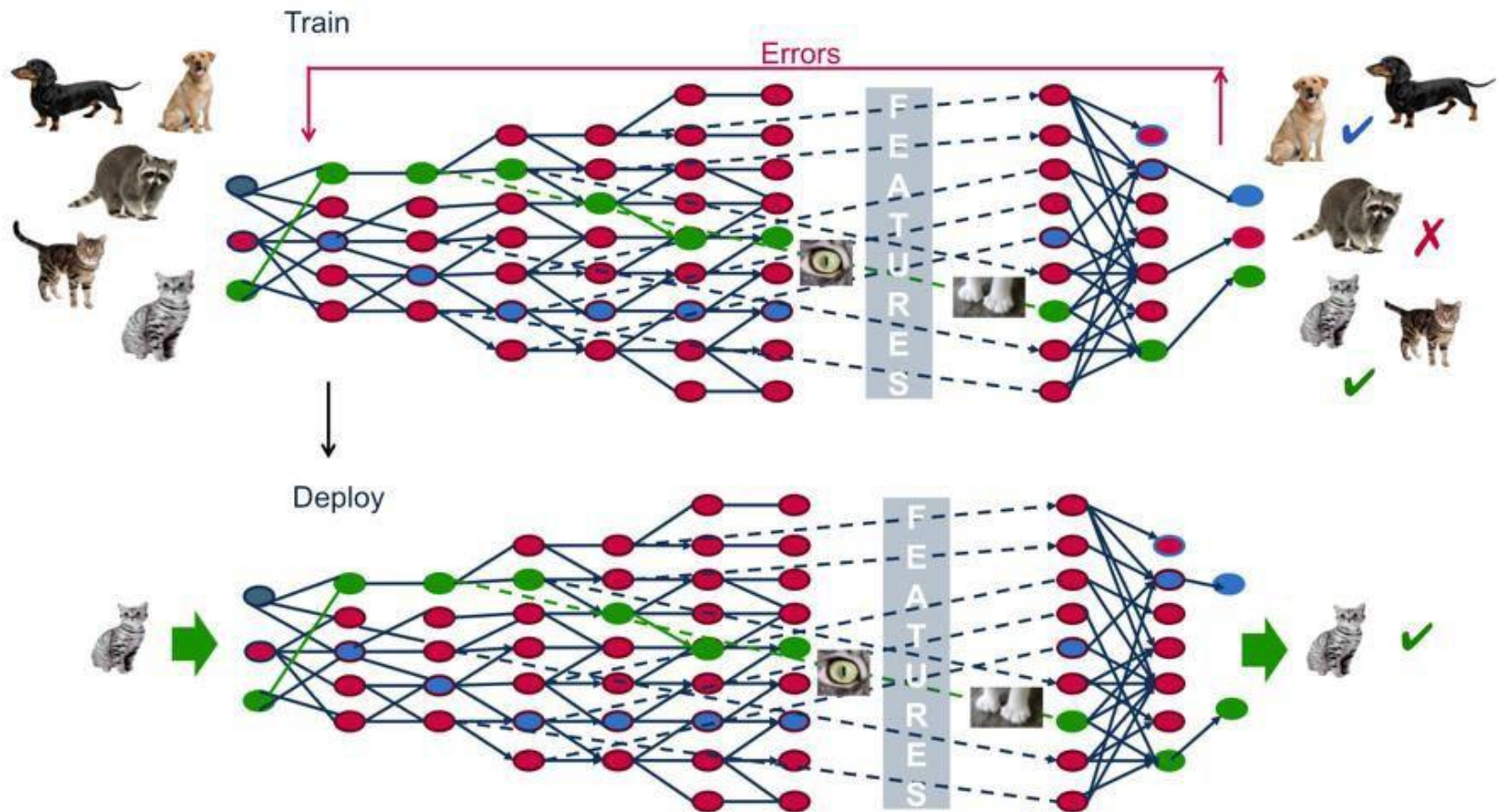
Business Applications of
Predictive Modeling at Scale -
KDD 2016 Tutorial

Overall Artificial Intelligence Journey



Data Protection & Information Governance

Deep Learning (multilayered neural networks)

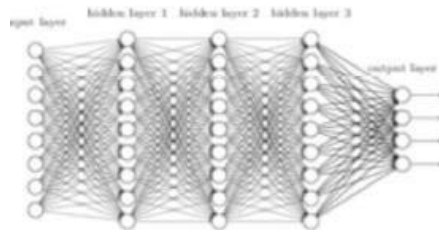


Data Protection & Information Governance

Deep Learning Algorithms

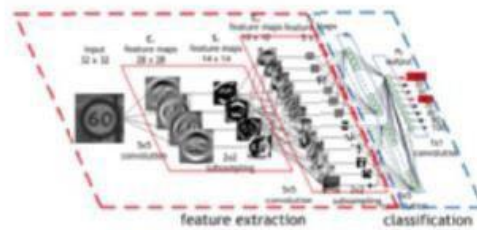
providing lift for
classification and
forecasting models

*Deep
Neural
Networks*



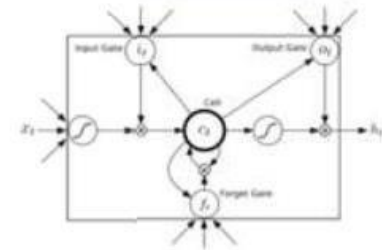
feature extraction
and classification of
images

*Convolutional
Neural
Networks*



for sequence of events,
language models, time
series, etc.

*Recurrent
Neural
Networks*



Data Protection & Information Governance

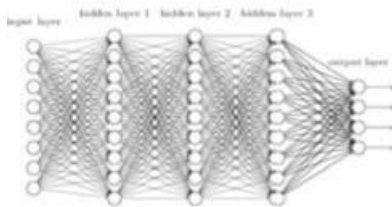
Deep Learning Algorithms

providing lift for
classification and
forecasting models

Deep
Neural
Networks

Deep neural networks for improved traditional algorithms.

- **Finance:** Enhanced fraud detection through identification of more complex patterns.
- **Manufacturing:** Enhanced identification of defects based on deeper anomaly detection.

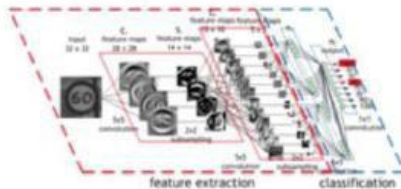


Data Protection & Information Governance

Deep Learning Algorithms

feature extraction
and classification of
images

*Convolutional
Neural
Networks*



Convolutional neural networks for images.

- **Retail:** In-store activity analysis of video to measure traffic.
- **Satellite images:** Labeling terrain and classifying objects.
- **Automotive:** Recognition of roadways and obstacles.
- **Healthcare:** Diagnostic opportunities from x-rays, scans, etc.
- **Insurance:** Estimating claim severity based on photographs.

Big Data Tools and Algorithm Exploration

The Problem Space

Machines are '*unable*' to classify images by recognising patterns, generalise from prior knowledge, nor adapt to different image environments

Image classification is the task of taking an input image and outputting a class or a probability of classes that best describes the image (a cat, dog)



What We See

Humans recognise these dogs naturally, we've instinctively done this since we were a baby

```
08 02 22 97 38 15 00 40 00 75 04 05 07 78 52 12 50 77 91 08
49 49 99 40 17 81 18 57 60 87 17 40 98 43 69 48 04 56 62 00
81 49 31 73 55 79 14 29 93 71 40 67 53 88 30 03 49 13 36 65
52 70 95 23 04 60 11 42 69 24 68 56 01 32 56 71 37 02 36 91
22 31 16 71 51 67 63 89 41 92 36 54 22 40 40 28 66 33 13 80
24 47 32 60 99 03 45 02 44 75 33 53 78 36 84 20 35 17 12 50
32 98 81 28 64 23 67 10 26 38 40 67 59 54 70 66 18 38 64 70
67 26 20 68 02 62 12 20 95 63 94 39 63 08 40 91 66 49 94 21
24 55 58 05 66 73 99 26 97 17 78 78 94 83 14 88 34 89 63 72
21 36 23 09 75 00 76 44 20 45 35 14 00 61 33 97 34 31 33 95
78 17 53 28 22 75 31 67 15 94 03 80 04 62 16 14 09 53 56 92
16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 85 57
86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 58 51 54 17 58
19 80 81 68 05 94 47 69 28 73 92 13 86 52 17 77 04 89 55 40
04 52 08 83 97 35 99 16 07 97 57 32 16 26 26 79 33 27 98 66
88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 57 05 54
01 70 54 71 83 51 54 69 16 92 33 48 61 43 52 01 89 19 67 48
```

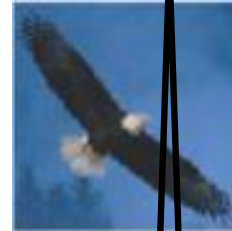
What Computers See

source /J. Seng GPU Teaching Kit is licensed by NVIDIA and New York University

Big Data Tools and Algorithm Exploration

Confusion Matrix: Measuring Success for Classification

Prediction:      

Image:      

True Positive True Negative False Negative False Positive True Negative True Positive

- True Positive: Correctly identified as relevant
- True Negative: Correctly identified as not relevant
- False Positive: Incorrectly labeled as relevant
- False Negative: Incorrectly labeled as not relevant

+ 'plus' sign is an image that is predicted to be a cat

– 'negative' sign is an image that is predicted to be NOT a cat

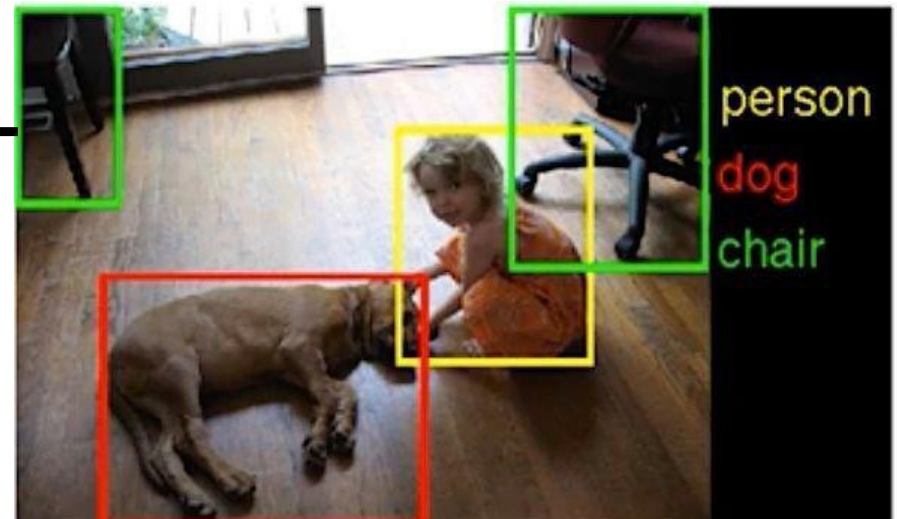
source /J. Seng GPU Teaching Kit is licensed by NVIDIA and New York University

Big Data Tools and Algorithm Exploration

Convolutional Neural Nets

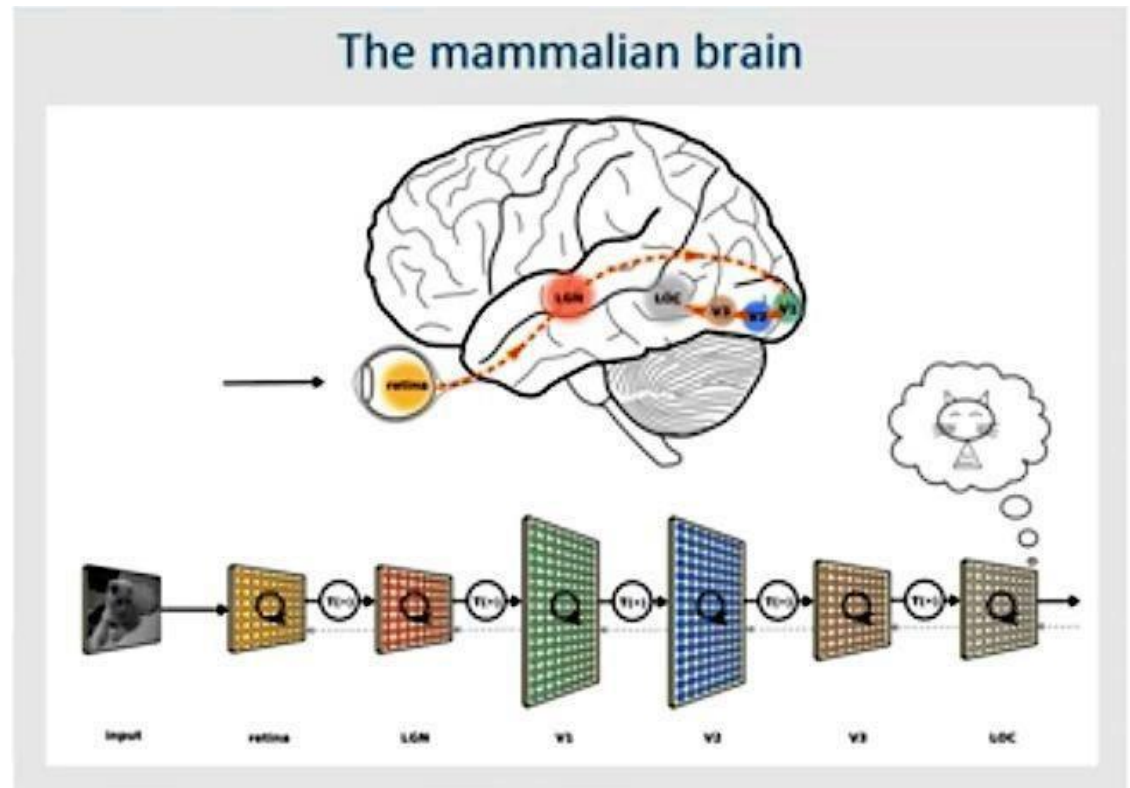
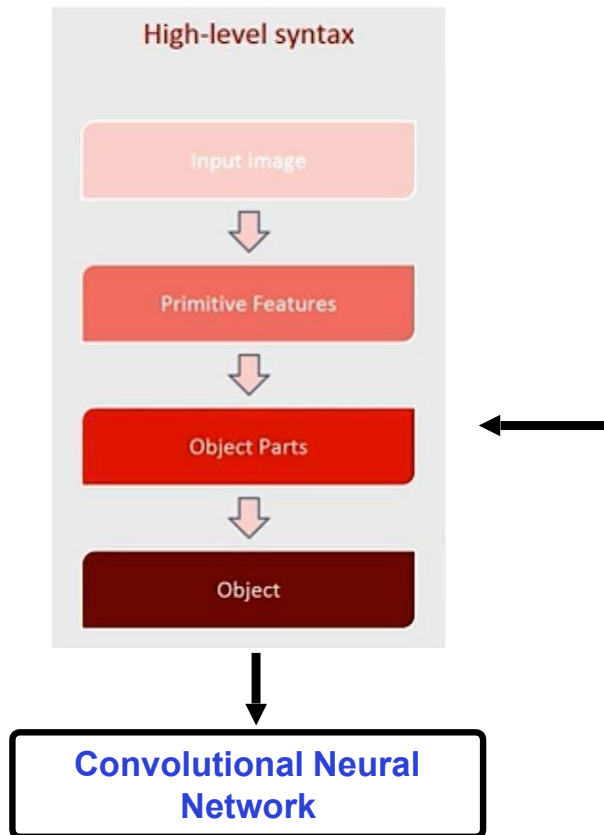
Computes the error derivative with respect to a weight for the output layer

- Signal and image processing
- Handwritten text/digits recognition
- Natural object classification (photos and videos)
- Segmentation
- Face detection
- Recommender systems
- Speech recognition
- Natural Language Processing



Big Data Tools and Algorithm Exploration

Convolutional Neural Nets

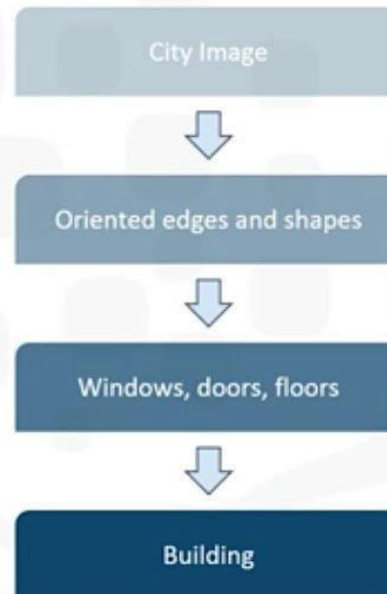


Big Data Tools and Algorithm Exploration

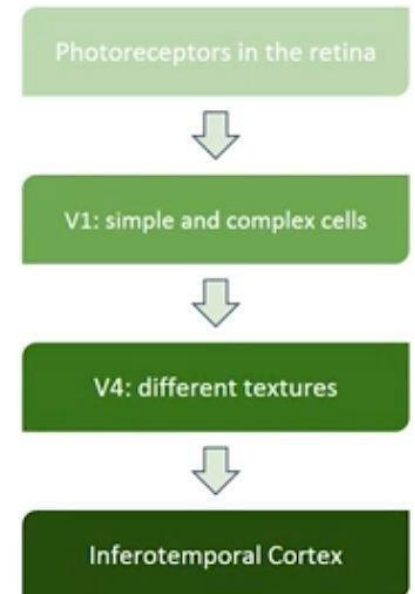
Convolutional Neural Nets



How the scene is organized

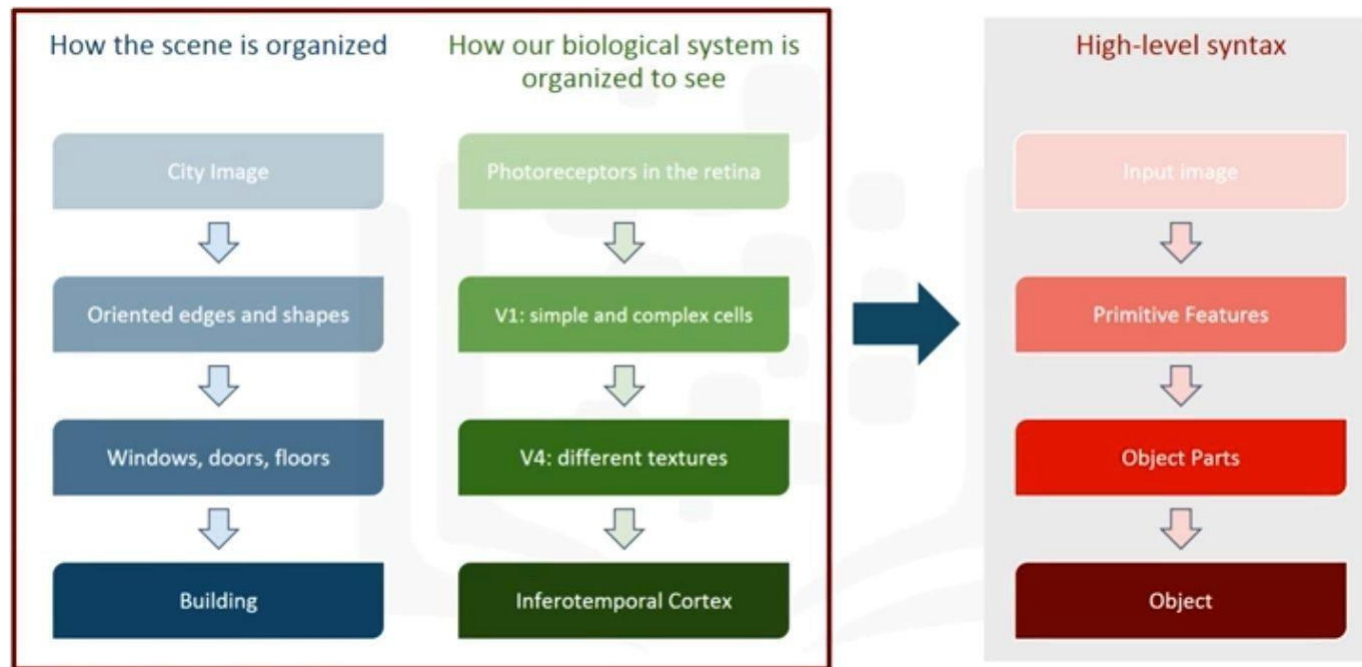


How our biological system is organized to see



Big Data Tools and Algorithm Exploration

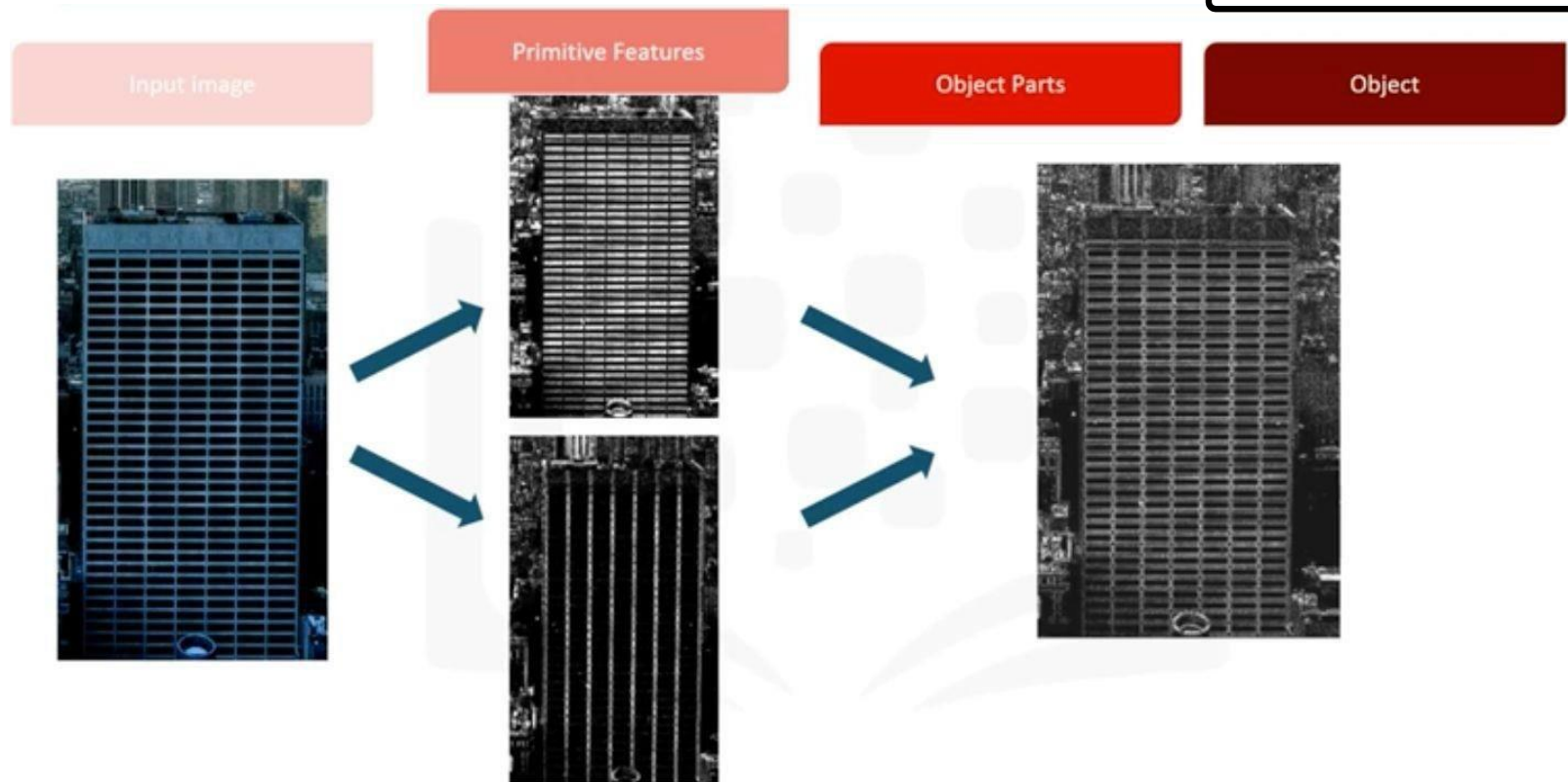
Convolutional Neural Nets



Big Data Tools and Algorithm Exploration

Convolutional Neural Nets

A Visual Explanation



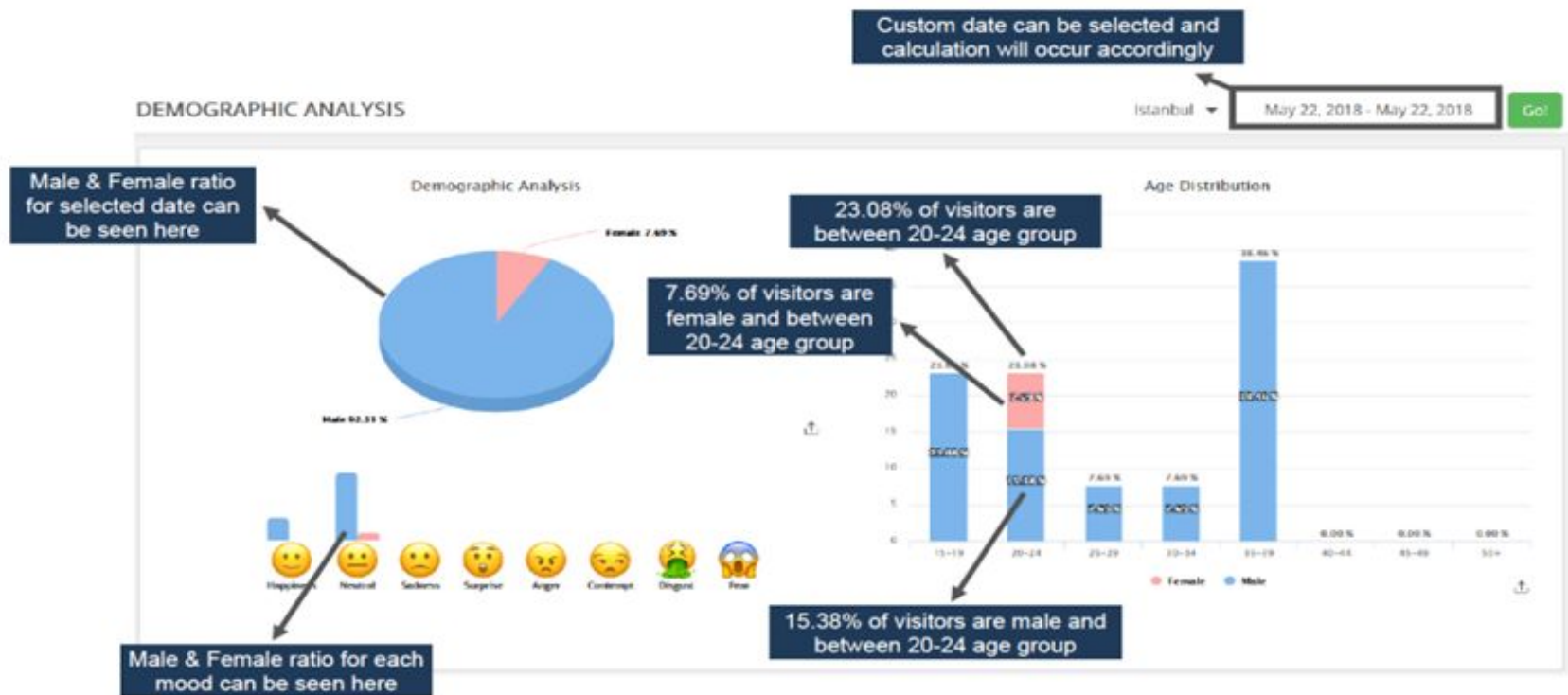
Deep Learning Case Study - Object Detection

- Identify which set of people like which see different item in the shop
- Identify which item do the customer mostly like



Deep Learning Case Study - Demographic Dashboard

Make use of facial recognition to detect Age Group, Gender, Emotion that demonstrate in dashboard



Location Tracking

- Customer locations can now be displayed on an shop floor plan in real time.
- Digital wayfinding with turn-by-turn navigation for your app, interactive kiosks and website.
- Conduct Data Analysis on how your customers are using your space.



Data Scientist Jobs



FIREBLAZE AI SCHOOL

Data Scientist

Also known as Data Manager,
Statistician

• Tools that need to be mastered •



Python



R Programming



SQL

• Skills that need to be mastered •



Programming



Statistics



Machine Learning



Data Visualization



FIREBLAZE AI SCHOOL

Data Engineers

Also known as Data Architects

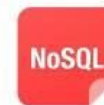
• Tools that need to be mastered •



Python



hadoop



NoSQL

• Skills that need to be mastered •



Programming



Data Mining

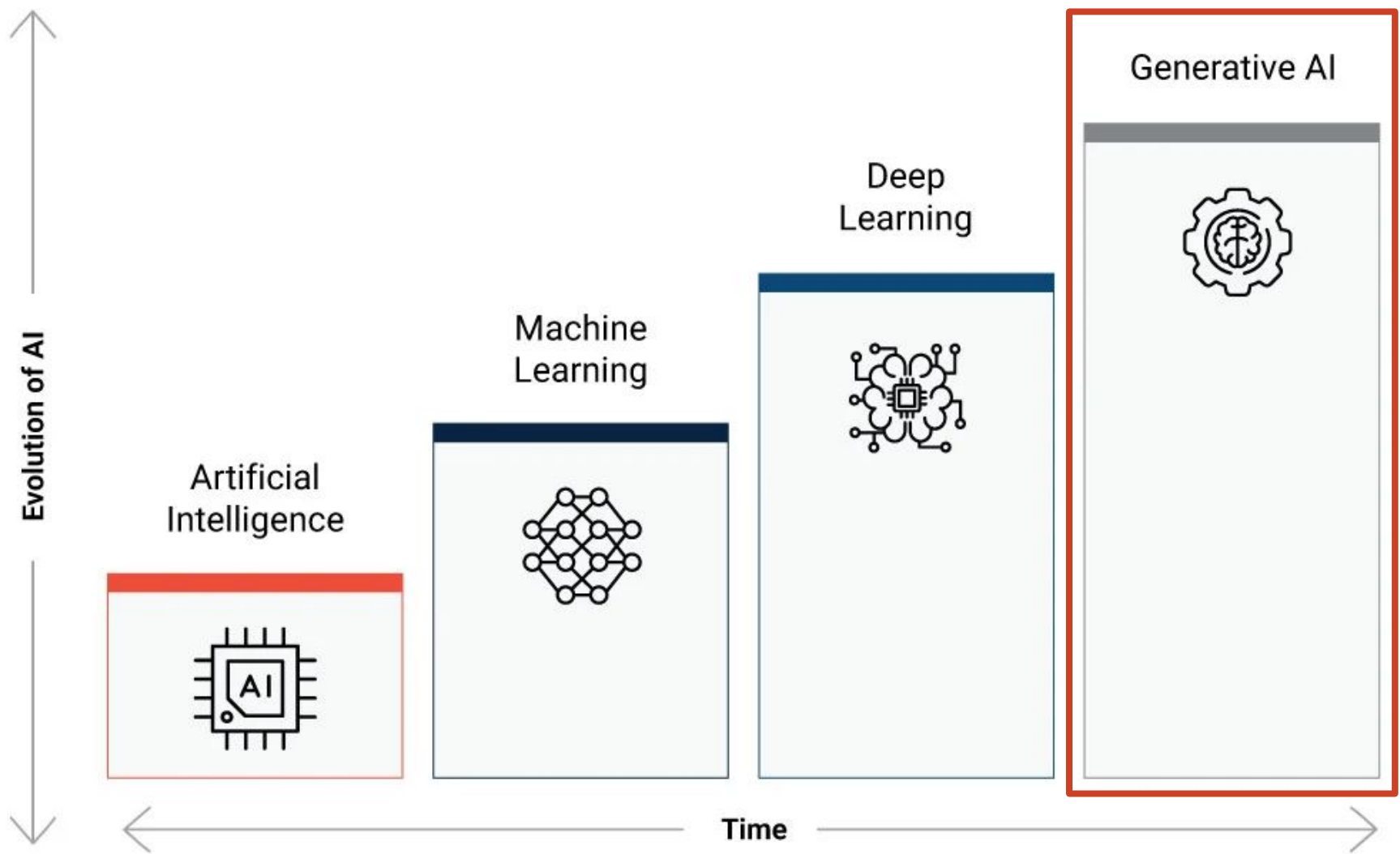


Database
architecture



Statistical modeling &
regression analysis

Overall Artificial Intelligence Journey



Generative AI

Generative AI refers to models that can create new text, code, images, audio, or video based on patterns learned from large datasets

Common underlying models are large language models (LLMs) for text/code, and diffusion or transformer-based models for images and video.

Major types of generative AI tools:

- Text & chat: ChatGPT, Claude, Gemini, Microsoft Copilot
- Coding assistants: GitHub Copilot
- Image & video: DALL·E, Midjourney, Synthesia for generating images, graphics
-

Generative AI using Python – GPT2

Open Colab Link at

https://colab.research.google.com/drive/1E6Hdzvp0IgUI1cG_Ddd0FCYQ9-FG7OJ2

```
prompt : write a 10 words with 9 letters
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results.

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

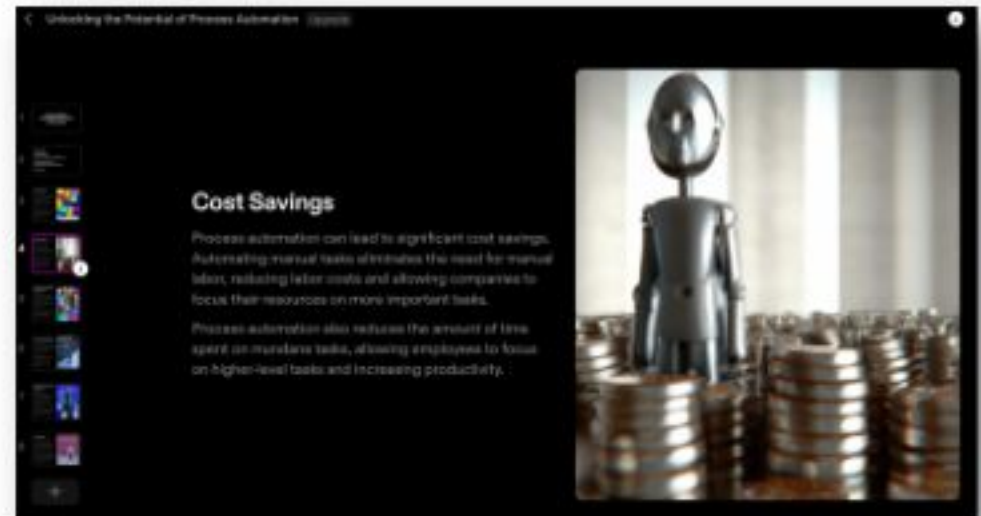
Introduction to Business Process Automation

AI Applications – Tome

- Input → Presentation topic
- Output → Slides with outlines and artworks
- <https://tome.app/>

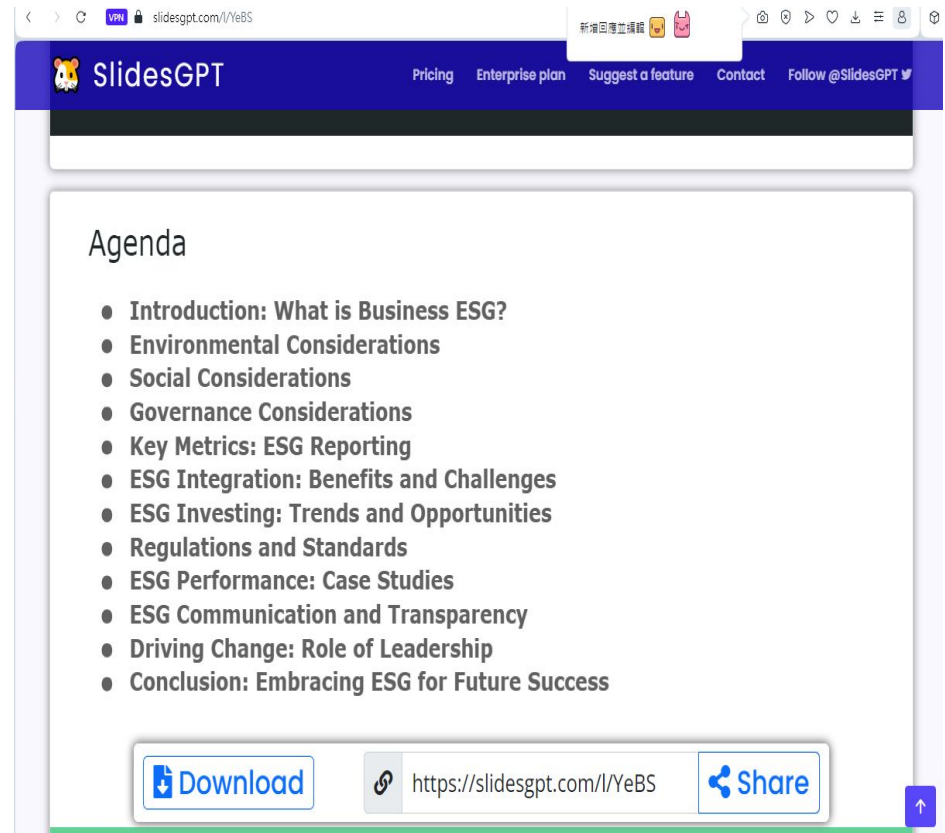
Unlocking the Potential of Process Automation

Introduction
Cost Savings
Improved Customer Experience
Increased Efficiency
Data Collection and Analysis
Conclusion



PPT AI Generator – SlideGPT

1. Open Browser and enter <https://slidesgpt.com>
2. Enter Prompt "I want to create a deck that includes Business ESG"



<https://slidesgpt.com/I/YeBS>

PPT AI Generator – SlideGPT

What is Business ESG?

ESG, Environmental, Social, Governance, Definition

- **ESG Framework:** ESG refers to Environmental, Social, and Governance factors that measure the sustainability and ethical impact of an investment in a business or industry.
- **Environmental Considerations:** The environmental aspect of ESG encompasses the impact of a company's operations on the environment, including resource usage, emissions, and waste management.
- **Social Considerations:** The social aspect of ESG evaluates a company's relationships with its employees, stakeholders, and the communities in which it operates, focusing on diversity, labor practices, and human rights.
- **Governance Considerations:** The governance aspect of ESG assesses the internal systems and controls of a company, including leadership, executive pay, shareholder rights, and ethical business practices.



Photo by Omar Quazi on Unsplash

Environmental Considerations

Environmental Impact, Climate Change, Carbon Footprint, Renewable Energy

- **Environmental Impact:** Assessing and mitigating the impacts of business activities on the natural environment, ecosystems, and resource depletion.
- **Climate Change:** Understanding and addressing the contribution of business activities to climate change through the reduction of greenhouse gas emissions and adaptation strategies.
- **Carbon Footprint:** Measuring and managing the total amount of greenhouse gases emitted directly or indirectly by a company, including Scope 1, 2, and 3 emissions.
- **Renewable Energy:** Exploring and investing in sustainable and renewable energy sources to reduce reliance on non-renewable resources and minimize environmental impact.



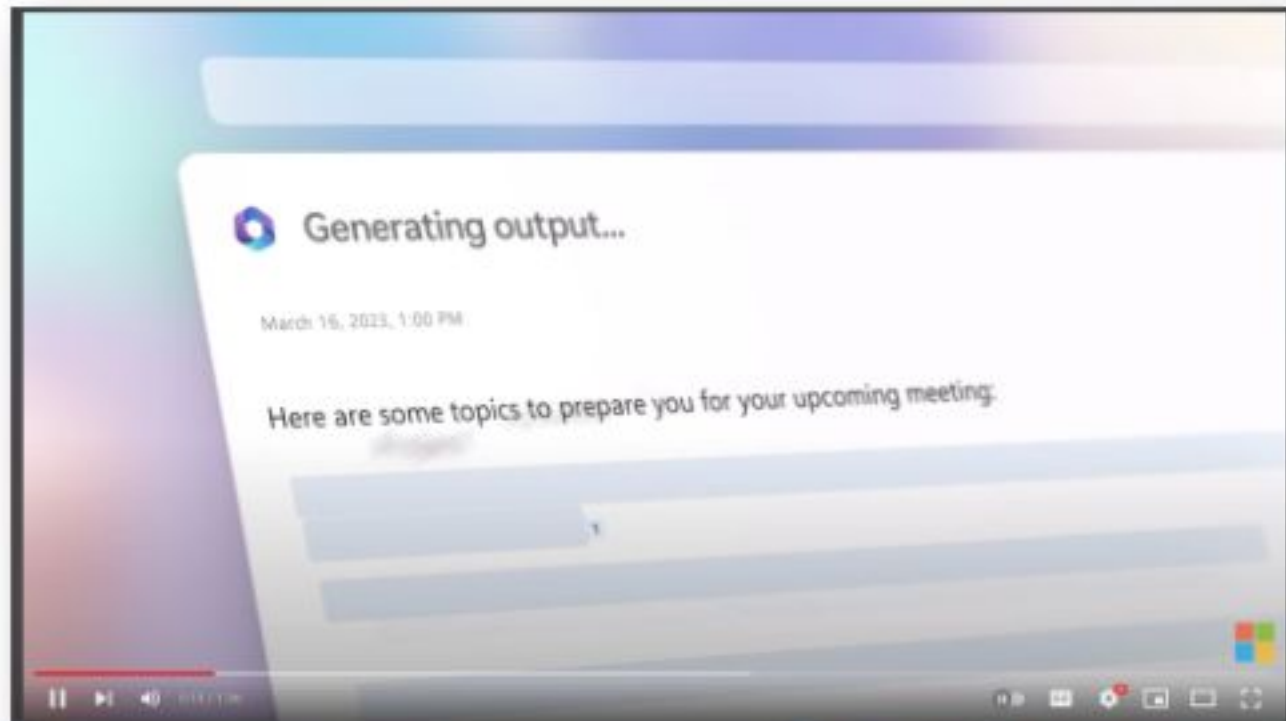
Photo by Tomasz Bazylinski on Unsplash

<https://slidesgpt.com/I/YeBS>

Introduction to Business Process Automation

AI Applications – Microsoft 365 Copilot

- Virtual Assistant
 - Word / Excel / PowerPoint / Outlook / Teams
- Ad: <https://youtu.be/S7xTBa93TX8>



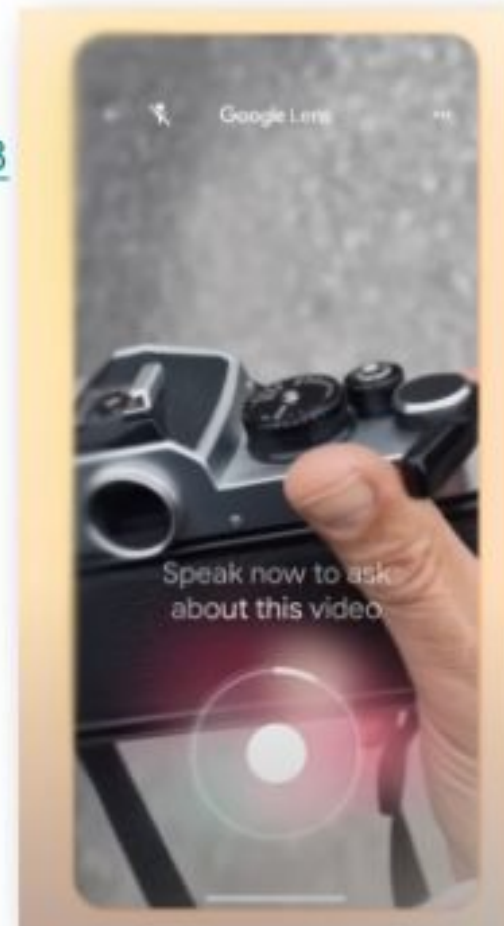
Introduction to Business Process Automation

AI Applications – Gemini

- Geolocation + Image/Video as Inputs

- Ad:

<https://youtu.be/s4InWsd-J6g?si=rwCwgMyHwUb40GAY&t=58>



Partners choose Microsoft for Data & AI

Accelerate time to value
with best in class platform & services



Pretrained AI
Services



Global Data
Services



Comprehensive
Platform

Innovate with AI everywhere –
in the cloud, at edge and on-premises



Cloud



Edge



On-premises

Use any language, any development
tool and any framework



PYTORCH



TensorFlow



Benefit from industry-leading security, privacy,
compliance, transparency, and AI ethics standards

>90% of Fortune 500 companies
use Microsoft Cloud

AI “Accelerators” – Cognitive Services

AI “Accelerators”
Solution specific
AI services
and patterns

[Azure Bot Service](#)
[Cognitive Services](#)

Vision 	Speech 	Language 	Knowledge 											
 What is in the image or video? Intelligent Image insights  <table border="1"> <tr> <td>Category</td> <td>People; 5 faces</td> </tr> <tr> <td>Adult/Racy?</td> <td>False/False</td> </tr> <tr> <td>Dominant colors</td> <td></td> </tr> <tr> <td>Accent color</td> <td></td> </tr> </table> Computer Vision	Category	People; 5 faces	Adult/Racy?	False/False	Dominant colors		Accent color		 Give me directions to the nearest local branch Speech to text  <table border="1"> <tr> <td>Convert spoken audio to text</td> </tr> <tr> <td>Convert text to spoken audio</td> </tr> <tr> <td>Extract intent of user</td> </tr> </table> Speech Service	Convert spoken audio to text	Convert text to spoken audio	Extract intent of user	 Play today's customer call recording Natural Language Processing Intent: PlayCall Content: Customer# DateTime.date: today  Now Playing 11/29/2016 Customer Call Language Understanding	 QnA Pair of this site? Automatic extraction of questions and answers What are your hours today? Today we are open from 7:00 AM to 10:00 PM. Do you have vegetarian options? Yes, we have vegetarian options available. QnA Maker
Category	People; 5 faces													
Adult/Racy?	False/False													
Dominant colors														
Accent color														
Convert spoken audio to text														
Convert text to spoken audio														
Extract intent of user														

AI “Accelerators” – Cognitive Services

Example: Industry leading neural text to speech

AI “Accelerators”

Solution
specific
AI services
and patterns

Azure Bot Service

[Cognitive Services](#)

Azure AI is built with a focus on customer centricity. We don't pretend to beat champions at games or design toys to attract attention. We build technology to help customers leverage AI to create life-changing experiences. If you've paid proper attention to this talk, you know how to leverage Azure AI to drive more ACR. But wait, there's more to come.



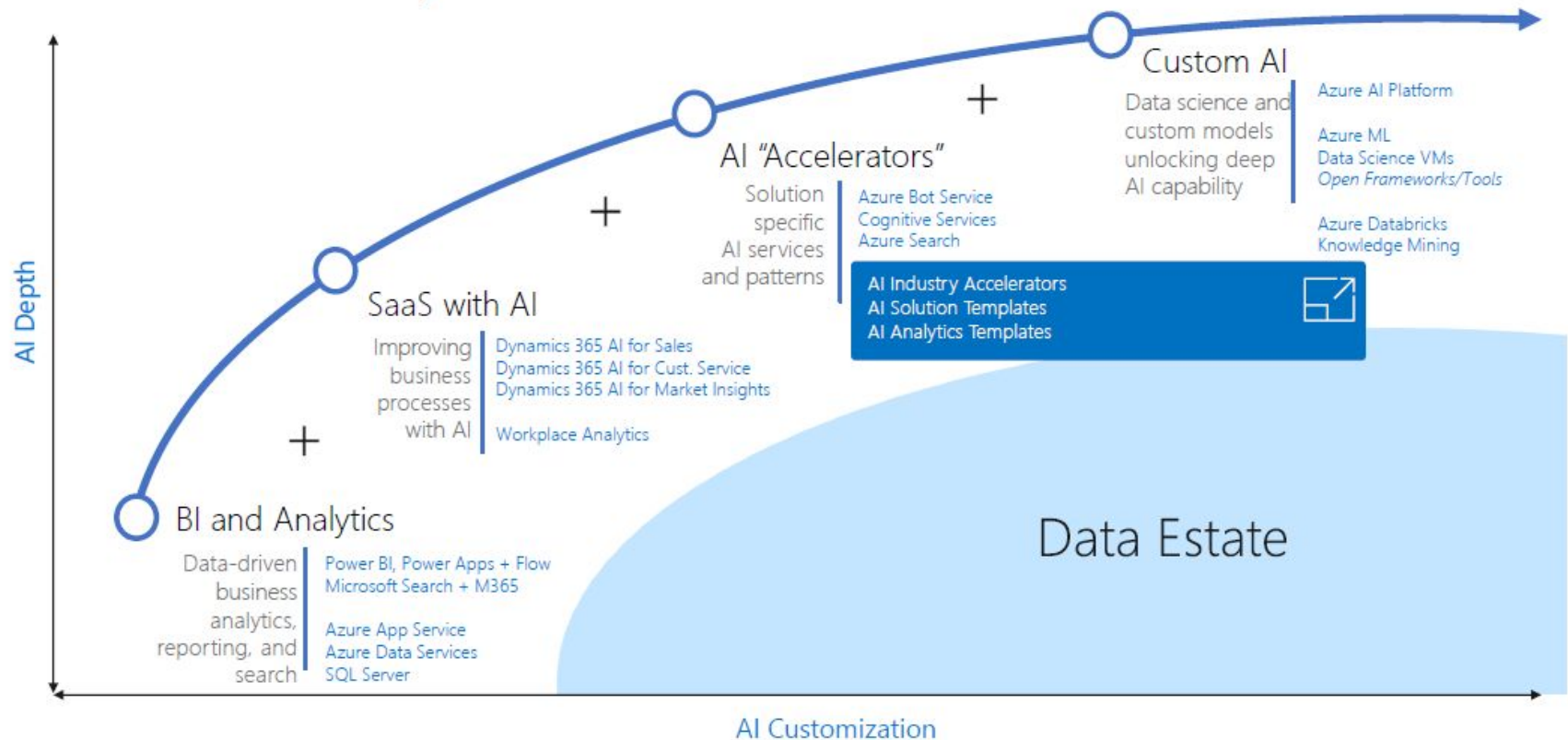
Sample 1



Sample 2

➤ Get started today: <http://aka.ms/NeuralTTSPreview>

The AI Journey – Where to Start



Microsoft AI Platform



Azure AI Services

PRE-BUILT AI

Cognitive Services

CONVERSATIONAL AI

Bot Service

CUSTOM AI

Azure Machine Learning



Azure Infrastructure

AI ON DATA

Cosmos DB

SQL DB

SQL DW

Data Lake

Spark ADB

DSVM

Batch AI

ACS

IoT Edge

AI COMPUTE

CPU, FPGA, GPU



Tools

CODING & MANAGEMENT TOOLS

VS Tools for AI

Azure ML Studio

Azure ML Workbench

Others (PyCharm, Jupyter Notebooks...)

DEEP LEARNING FRAMEWORKS

Third-party

Cognitive Toolkit

TensorFlow

Caffe

Others (Scikit-learn, MXNet, Keras, Chainer, Glueviz...)

Microsoft AI Platform



Azure AI Services

PRE-BUILT AI

Cognitive Services

CONVERSATIONAL AI

Bot Service

CUSTOM AI

Azure Machine Learning



Azure Infrastructure

AI ON DATA

Cosmos DB

SQL DB

SQL DW

Data Lake

Spark ADB

DSVM

Batch AI

ACS

IoT Edge

AI COMPUTE

CPU, FPGA, GPU



Tools

CODING & MANAGEMENT TOOLS

VS Tools for AI

Azure ML Studio

Azure ML Workbench

Others (PyCharm, Jupyter Notebooks...)

DEEP LEARNING FRAMEWORKS

Third-party

Cognitive Toolkit

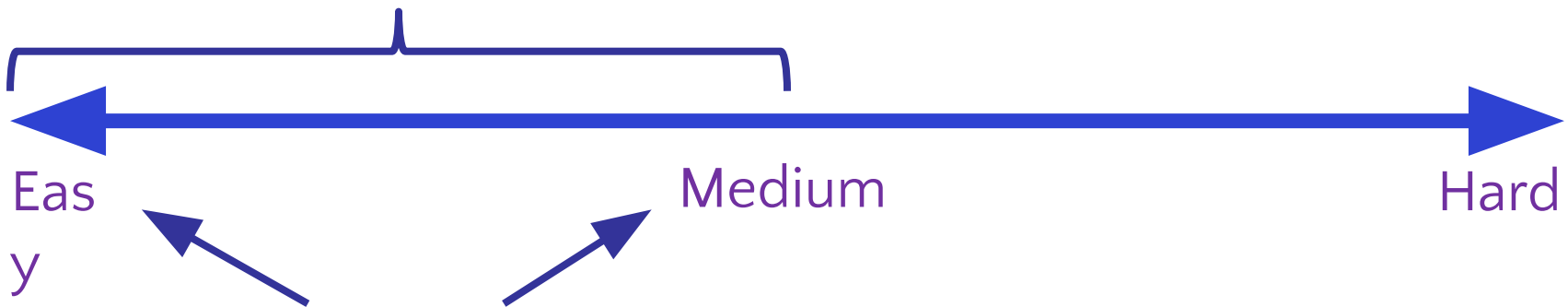
TensorFlow

Caffe

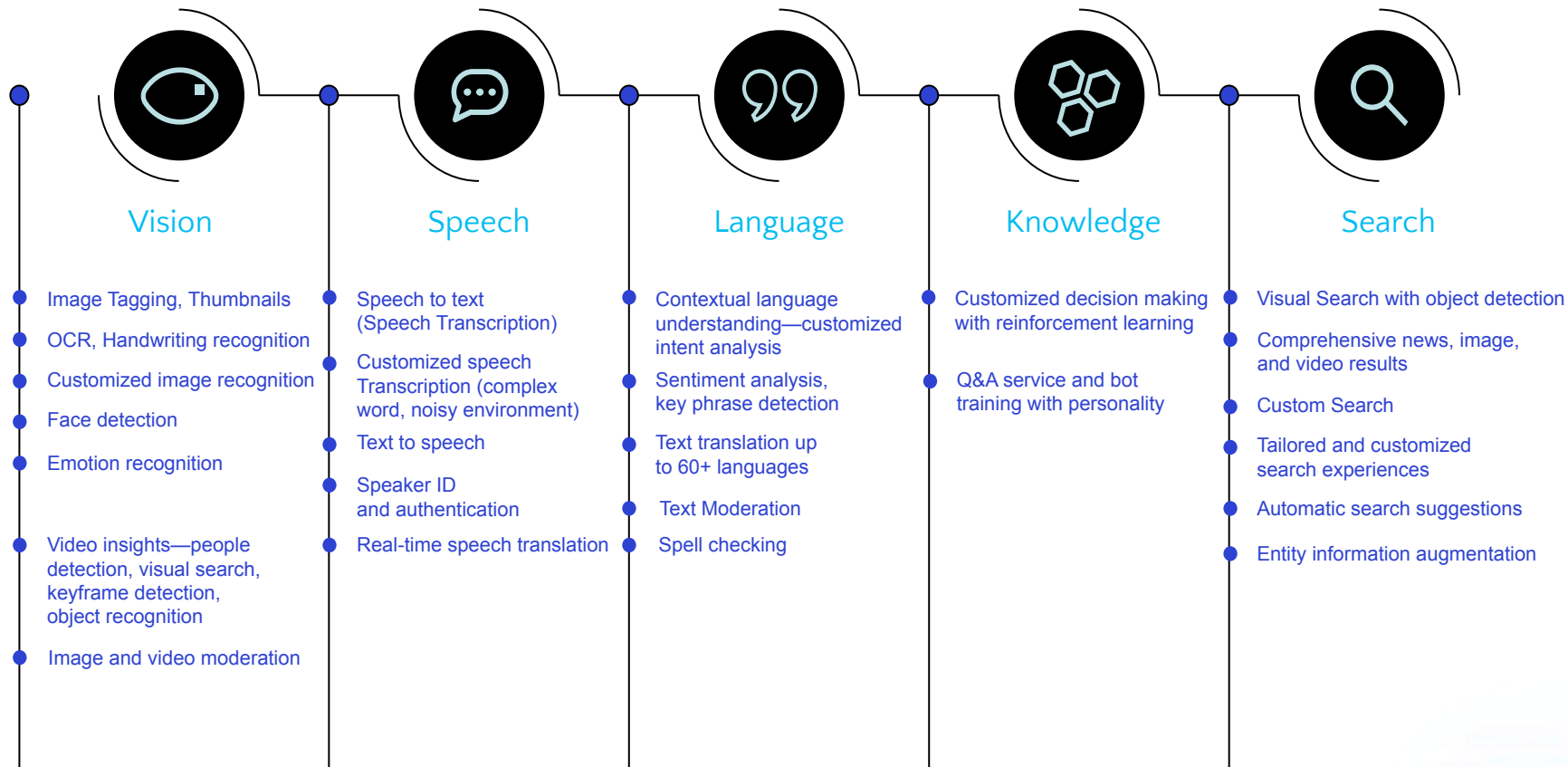
Others (Scikit-learn, MXNet, Keras, Chainer, Glueviz...)

Azure Cognitive Services

A set of **simple APIs** that harness **Machine Learning**
so you can add AI (without needing a deep understanding of AI)



Azure Cognitive Services



Computer Vision



FEATURE NAME:	VALUE
Description	{ "tags": ["indoor", "table", "sitting", "black", "pair", "laying", "white", "remote", "sink"], "captions": [{ "text": "a pair of black headphones on a table", "confidence": 0.307312727 }] }
Tags	[{ "name": "indoor", "confidence": 0.9374284 }, { "name": "tool", "confidence": 0.9374284 }, { "name": "brush", "confidence": 0.4960692 }, { "name": "design", "confidence": 0.4172834 }, { "name": "key", "confidence": 0.29472214 }, { "name": "artisan", "confidence": 0.136394173 }, { "name": "knife", "confidence": 0.1352209 }]

Custom AI – Knowledge Mining

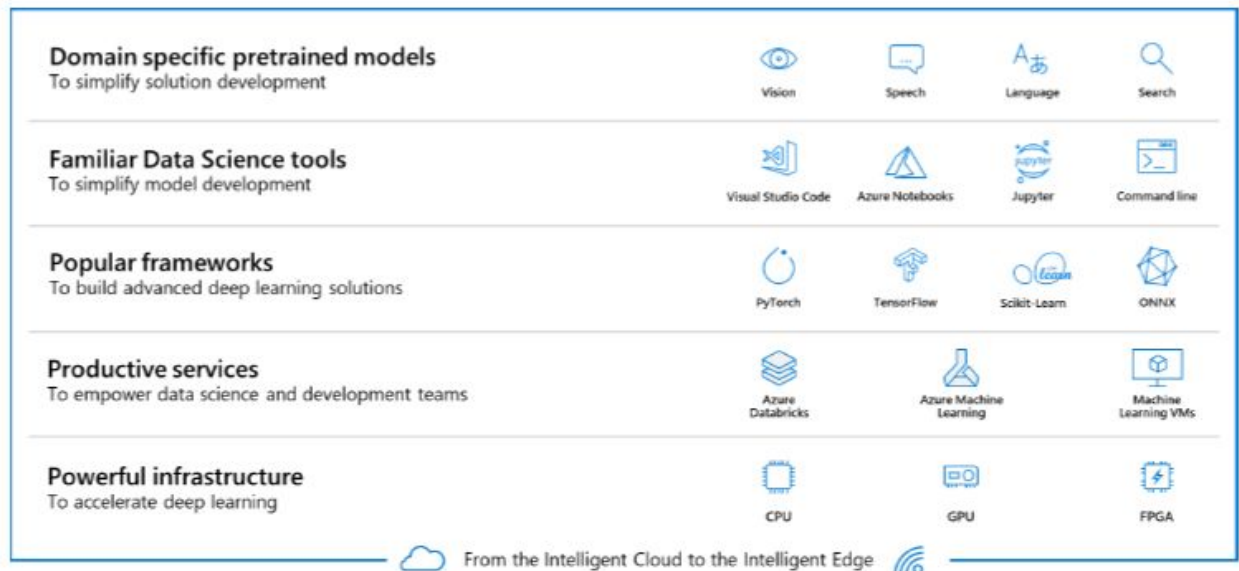
Custom AI

Data science and custom models unlocking deep AI capability

[Azure AI Platform](#)

[Azure ML](#)

[Knowledge Mining](#)



Custom AI – Azure ML

Custom AI

Data science and custom models unlocking deep AI capability

Azure AI Platform

[Azure ML](#)

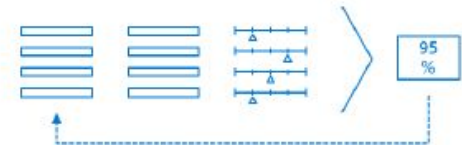
Knowledge Mining

Machine learning DevOps



Azure DevOps integration for CI/CD

Automated machine learning



Accelerated model building

Custom AI – Azure ML

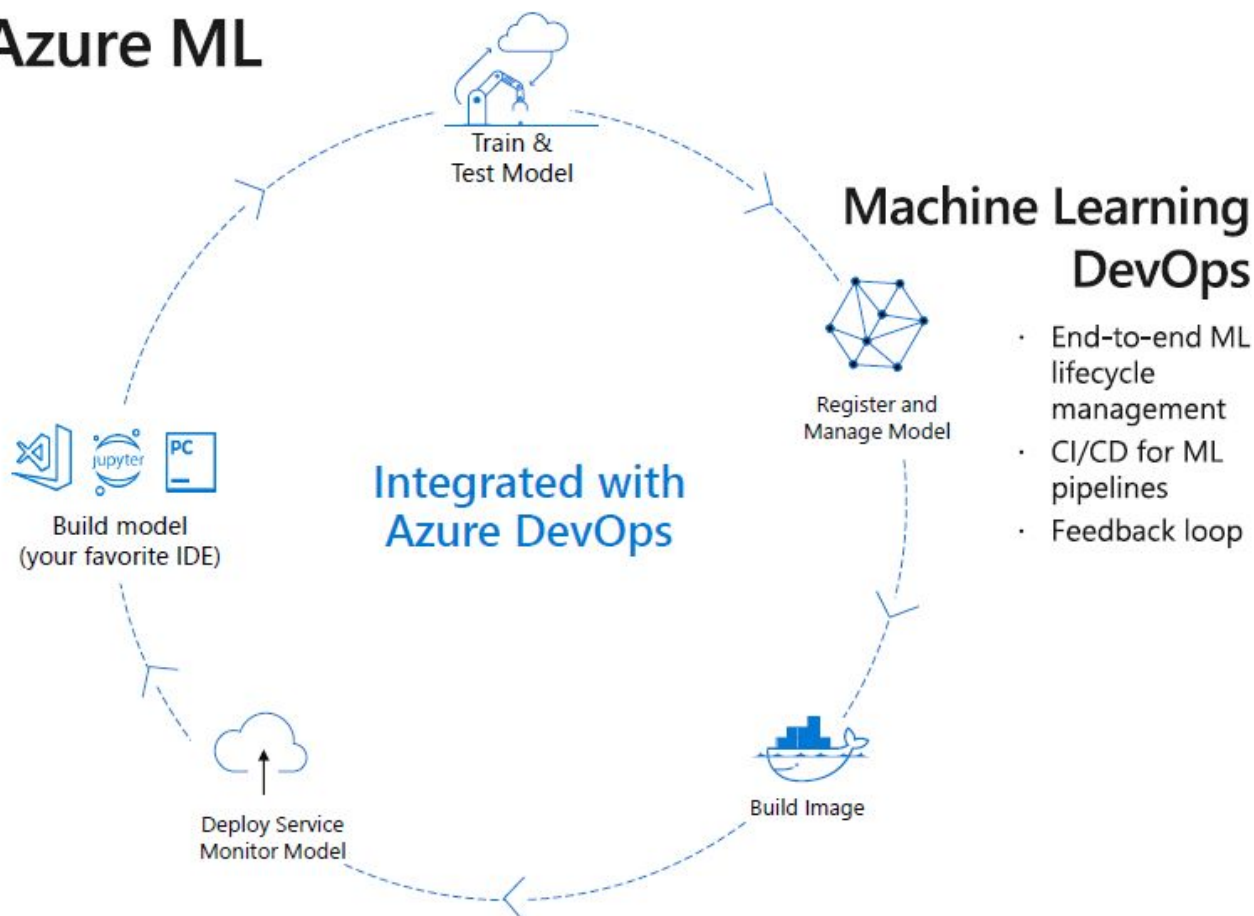
Custom AI– Azure ML

Custom AI
Data science and
custom models
unlocking deep
AI capability

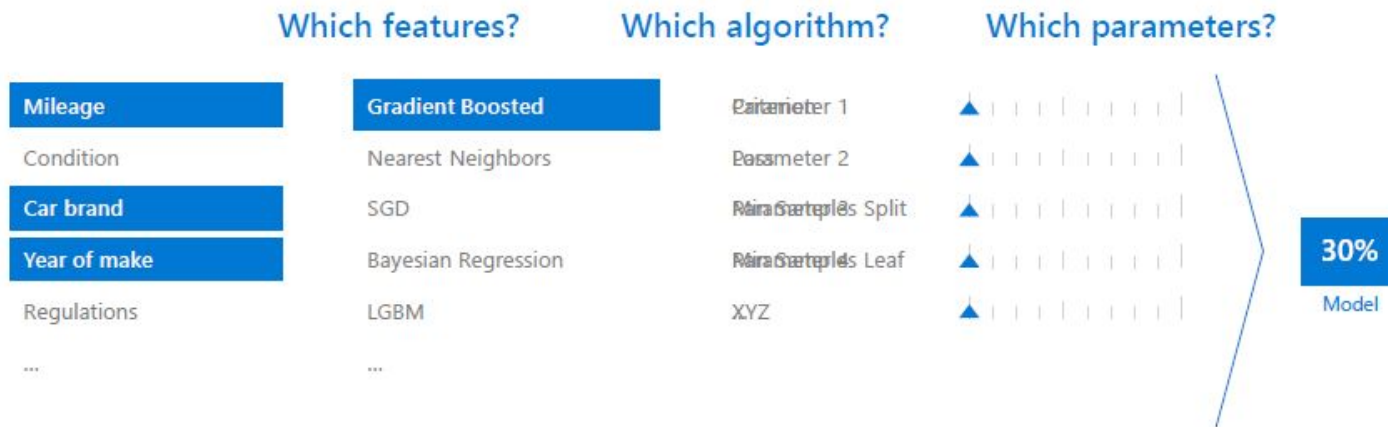
[Azure AI Platform](#)

[Azure ML](#)

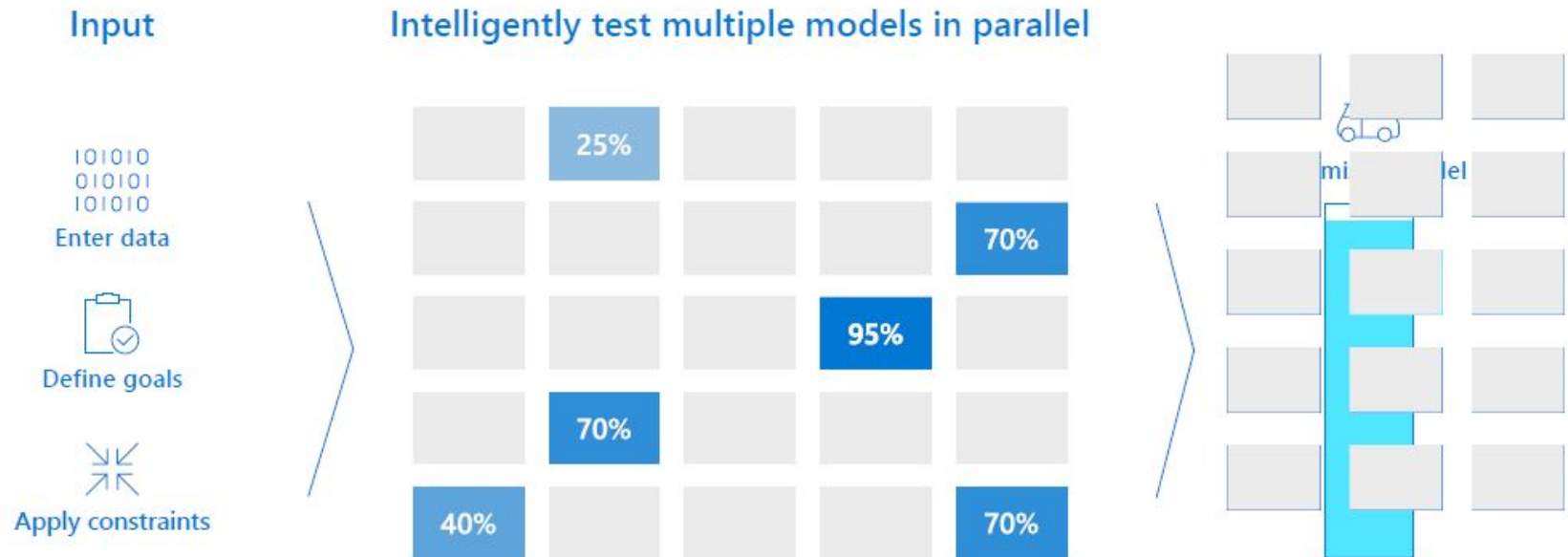
Knowledge Mining



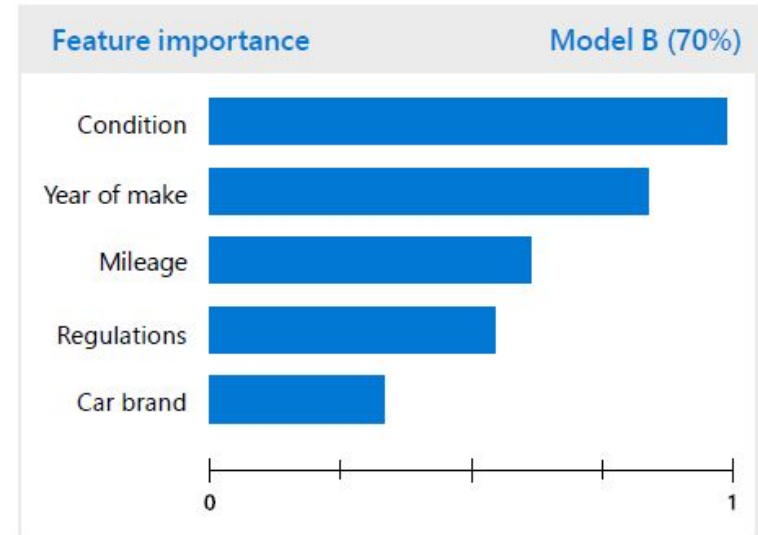
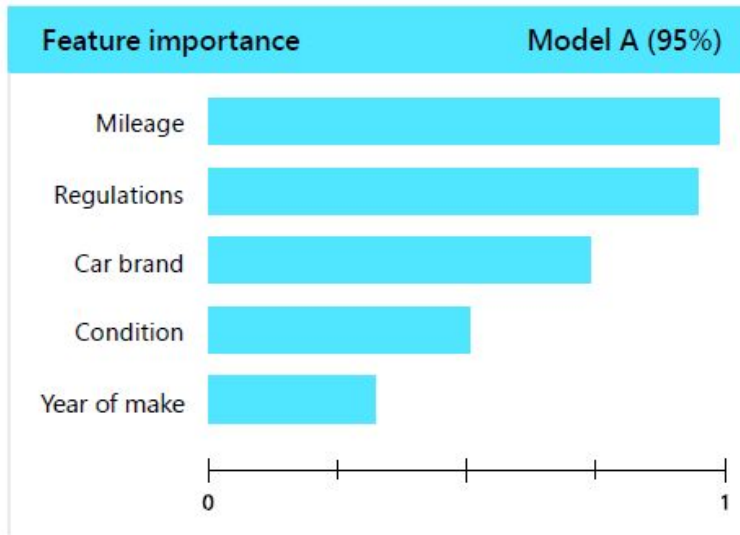
Model Creation is typically a time consuming process



Azure Machine Learning accelerates model development with automated machine learning



Azure Machine Learning accelerates model section with model explainability



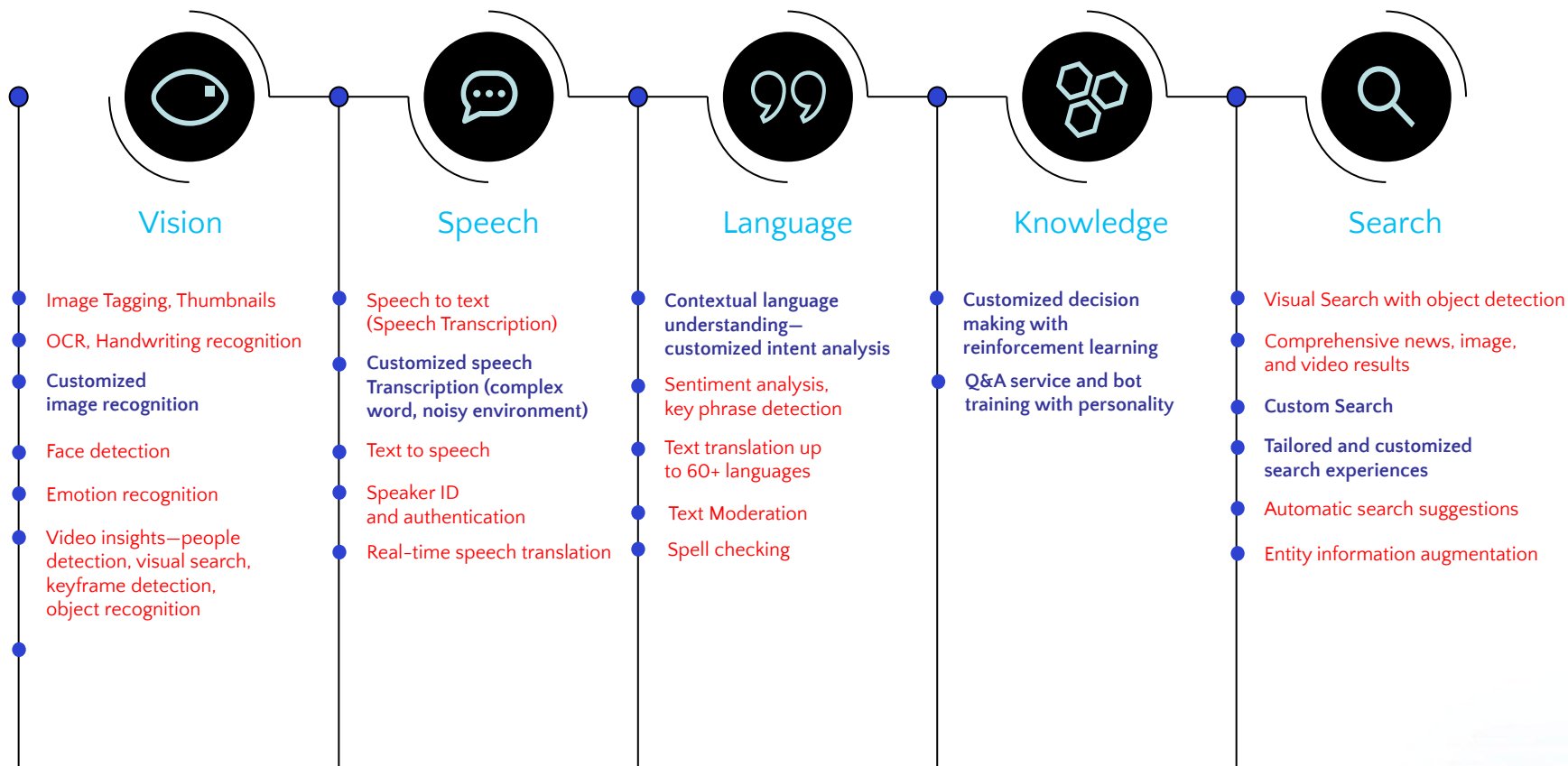
Bots Case Story – 1000+ Companies Engaging us

Scenario	Retail	Finance	Insurance	Telecoms	Government	Automotive	Manufacturing	Healthcare	Media	Events
Customer service	✓	✓	✓	✓	✓	✓	✓	✓		✓
Customer retail	✓	✓	✓	✓				✓		
Audio/speech analysis	✓	✓	✓	✓	✓				✓	
Translation		✓	✓							
Surveillance		✓			✓					
Knowledge extraction		✓	✓	✓			✓			
Video/photo analysis		✓			✓				✓	
Product identification	✓						✓	✓		
Digital assistant						✓				
Footfall analysis	✓									✓
HD maps and object detection						✓				

Bot Highlights – Improvements Everywhere



Azure Cognitive Services



Microsoft Learning - AI

<https://www.microsoft.com/en-us/ai/ai-school>

Conversational AI

Add intelligent chat to your apps and channels with AI-powered bots that incorporate features like natural language processing, intent recognition, and more.



Create bots with Azure Bot Service

Bots allow customers to interact with applications in a conversational way using text, graphics, or speech. From a simple question and answer dialog, to sophisticated pattern matching and state tracking, learn how to build a chat bot with QnA Maker and LUIS.

[Start the course >](#)



Create interactive conversational bots for Microsoft Teams

Learn how to create bots for custom Microsoft Teams apps, allowing users to interact with your web service through text, interactive cards, and task modules. Bots can be a few simple commands, or complex virtual assistants powered by AI and natural language.

[Start the course >](#)



Building bots with REST

Most Bot Framework bots are built using the Bot Framework SDK, which organizes your bot and handles all conversations for you. An alternative to the SDK is to send messages directly to the bot using a REST API. You can send and receive messages with users on any channel.

[Learn about building bots >](#)

Azure Cognitive Services

<https://docs.microsoft.com/en-us/learn/certifications/roles/ai-engineer>

AI Engineer certification path

The ai engineer certification path is organized into 3 levels: Fundamentals, Associate and Expert.

View by:

Azure AI Engineer ▾

An optional start for those new to Azure



FUNDAMENTALS CERTIFICATION

Microsoft Certified: Azure Fundamentals

Complete an associate certification



ASSOCIATE CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate

AI Engineer certifications

Explore ai engineer certifications most sought after by employers



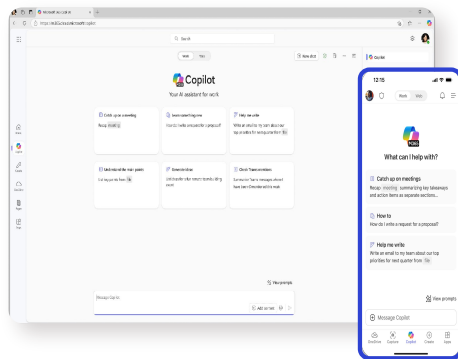
Microsoft Certified: Azure AI Engineer
Associate

Microsoft 365 Copilot Chat

Microsoft 365 Copilot Chat is at the center of Microsoft 365 Copilot.

Advantage:

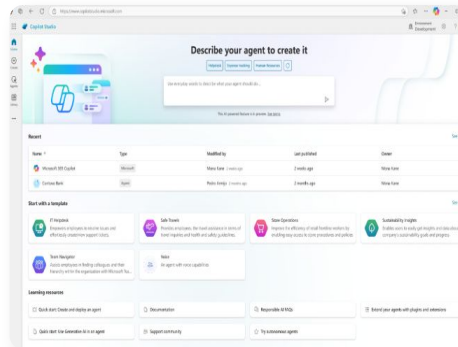
- Powerful : Create content and catch up quickly.
- Insightful : Ask questions to help you gather information.
- Goes where you go : Use Copilot Chat on the web, from the Copilot desktop app, Outlook, Microsoft Teams, PowerPoint, Word, Excel, OneNote, or on your mobile device.



Chat

Free AI chat -- powered by GPT 4o and grounded in the web

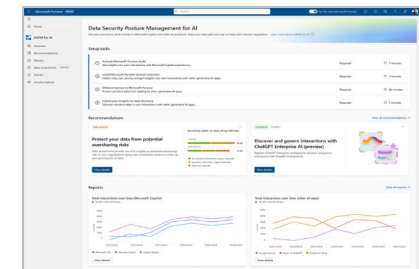
+



Agents

Integrated into the chat experience and paid for on a consumption basis

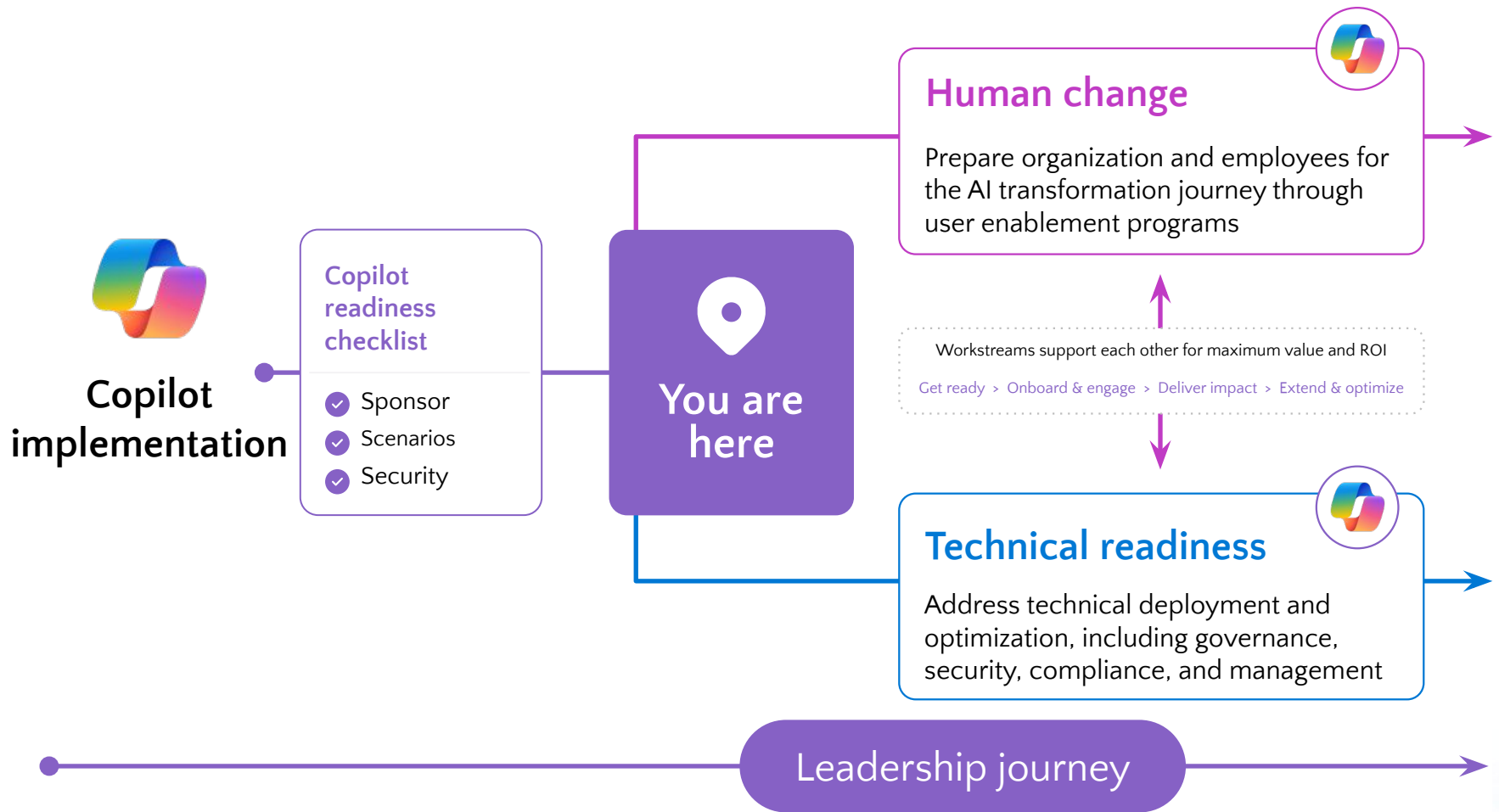
+



IT Controls

Enterprise Data Protection and management

Microsoft 365 Copilot Implementation



Accelerate your AI workforce transformation

Communications

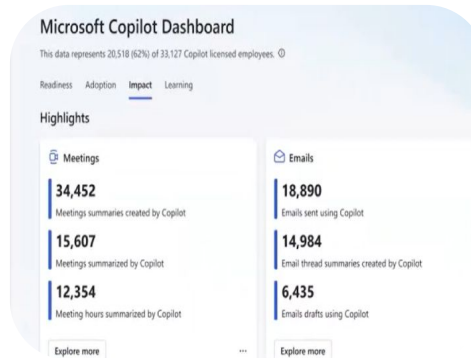


Copilot Communities to facilitate user enablement

- Share best practices
- Access company announcements
- Seek support from peers and IT

aka.ms/CopilotCommunities

Measurement

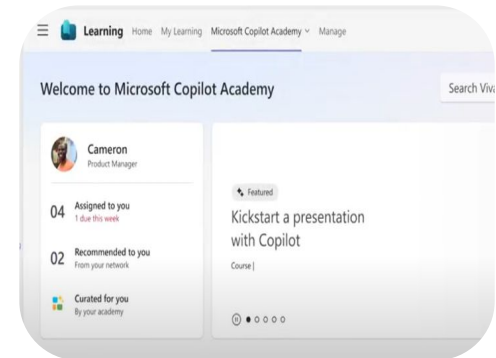


Copilot Analytics for ROI and impact assessment

- Copilot Dashboard for leaders
- Viva Insights for customizable analysis
- Copilot business impact reports

aka.ms/CopilotAnalytics
aka.ms/CopilotDashboard

Skilling



Copilot Academy for user skill development

- Curated learning paths
- Hands-on prompt guidance
- Content created by Microsoft experts

aka.ms/CopilotAcademy

Microsoft Copilot Studio

Build your own agents

Create and publish a custom agent for your organization using the intuitive building experience enhanced with large language models and generative AI

Customize Microsoft Copilot

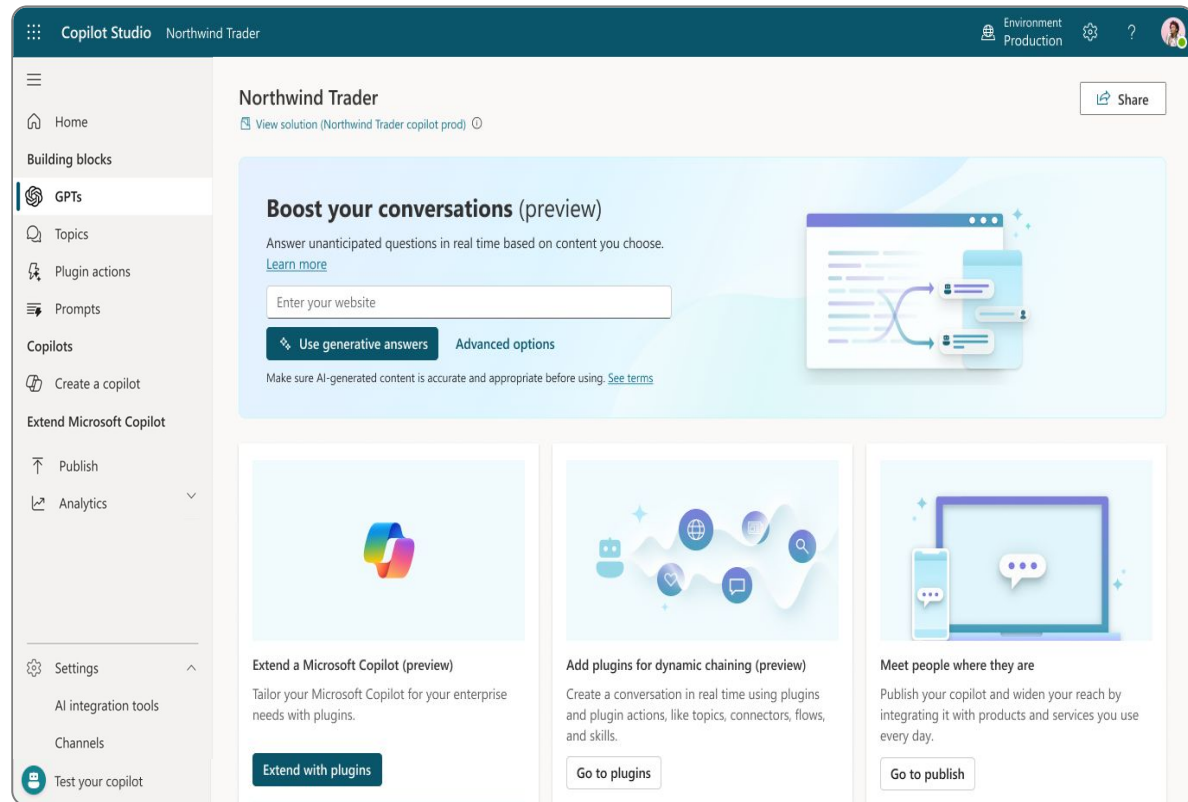
Extend and customize 1st party agents with your own enterprise scenarios. Copilot Studio will be included with the Microsoft 365 Copilot SKU.

Connected platform

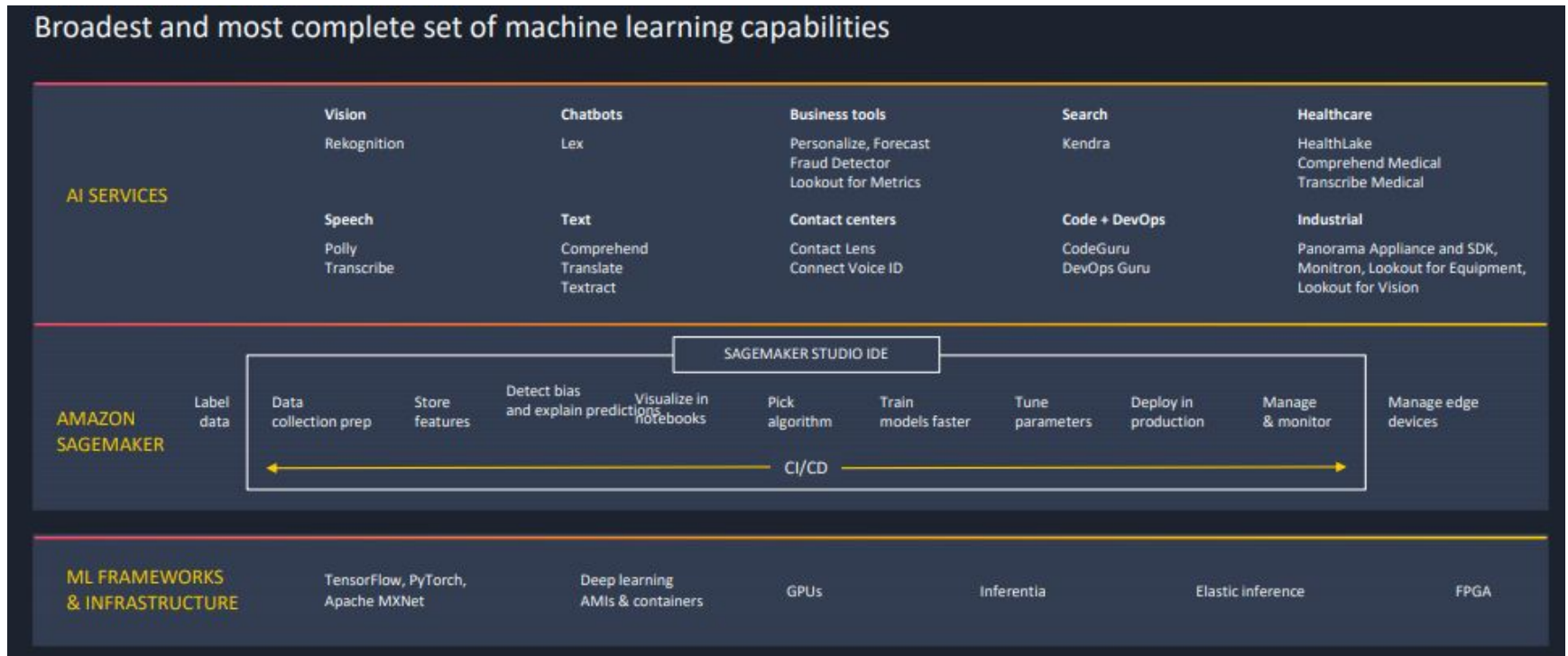
Integrates and exposes various Microsoft's conversational AI technology stacks – integrated with Azure AI Studio, Azure Cognitive Services, Azure Bot Framework, Power Platforms AI models, and more

Manage experiences

Governance and control features to monitor usage with full visibility of customizations, standalone agents as well as who is building and customizing them.



The AWS ML Stack





AWS AI and ML Services



AWS AI and ML Cases Study

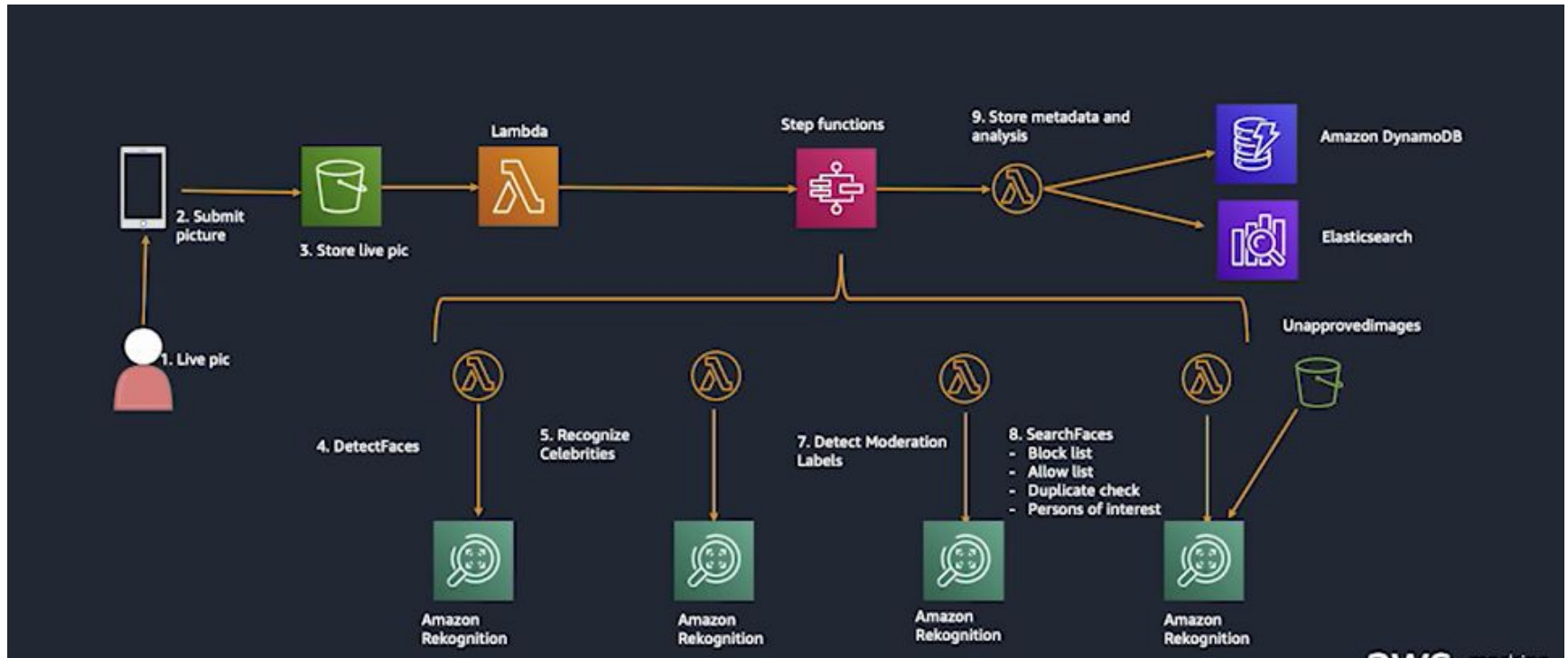
The screenshot displays the AWS Rekognition console interface. On the left, a navigation menu lists various services under 'Amazon Rekognition', with 'PPE detection' highlighted. The main area shows a video frame of a person wearing a blue surgical mask. Below the frame, the 'Summarization inputs' section includes checkboxes for 'Required PPE' (Face cover, Hand cover, Head cover) and a 'Required minimum confidence' of 80%. The 'Results' section on the right provides a summary of findings:

Summarization results	
Persons with required equipment (Ids):	[]
Persons without required equipment (Ids):	[]
Persons indeterminate (Ids):	[0]

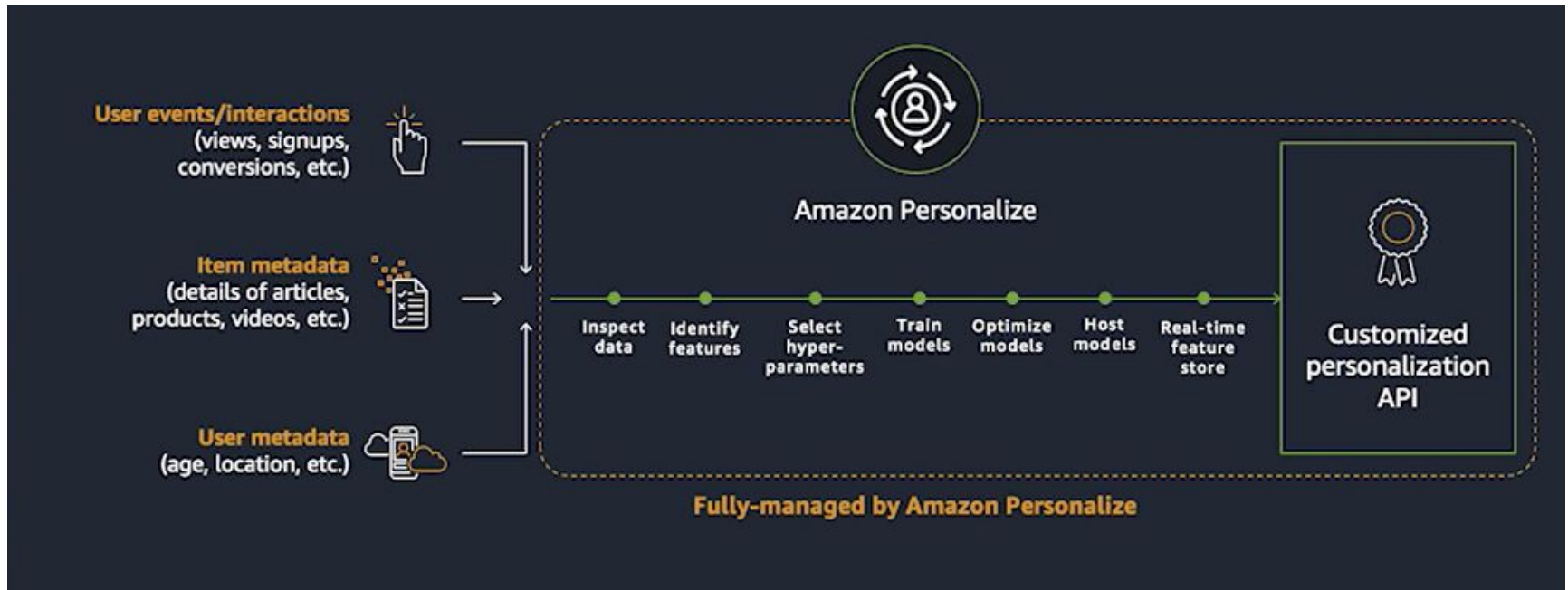
Per-person results	
Person ID: 0/0	
Person detected	99.9 %
Face detected	99.9 %
Face cover detected	99.9 %
Face cover on nose : false	50.2 %
Head detected	100 %

An orange arrow points to the 'Face cover on nose : false' result, indicating a specific detection detail.

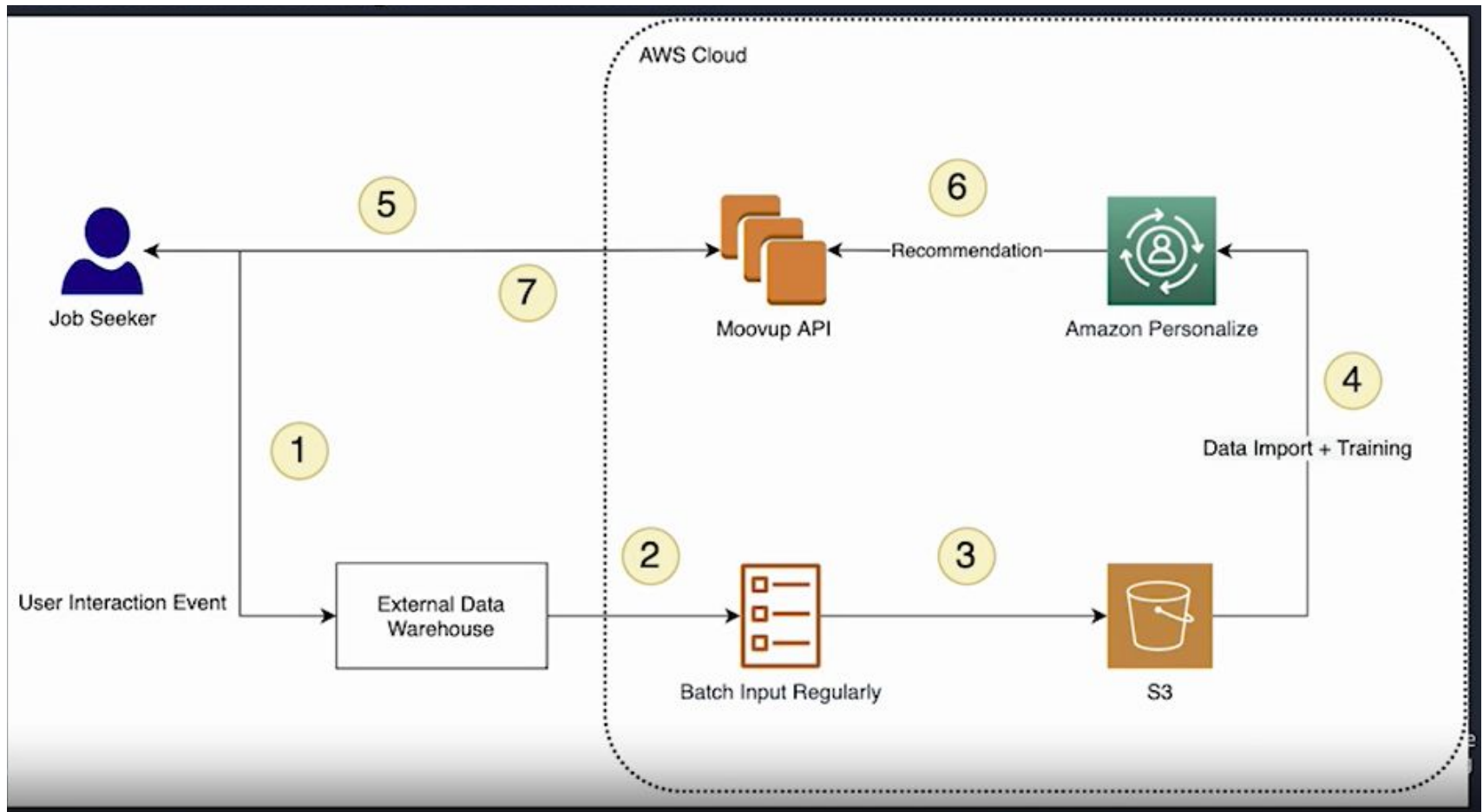
AWS Solution



HOW IT Works – Amazon Personalize



Recommendation System Architecture



Amazon Learning - AI

<https://aws.amazon.com/events/events-content/?awsf.filter-language=language%23english&awsf.filter-topic=event-topic%23ai-ml>

EMB037	LEVEL 200	EMB035	LEVEL 200	EMB034	LEVEL 200
					
[NEW LAUNCH] Understand ML model predictions...		[NEW LAUNCH] MLOps for edge devices with ...		[NEW LAUNCH!] Accelerate data preparation with...	
Machine learning (ML) models may generate predictions that are not fair, whether because of biased data, a model that contains bias, or bias that emerges over time as real-world conditions change. Likewise, closed-box ML models are opaque, making it		In this session, learn about Amazon SageMaker Edge Manager, a new capability of SageMaker that helps developers operate machine learning (ML) models on a fleet of edge devices, helping solve challenges with constraints and maintenance of ML		Preparing training data can be tedious. Amazon SageMaker Data Wrangler provides a faster, visual way to aggregate and prepare data for machine learning. In this session, learn how to use SageMaker Data Wrangler to connect to data sources and use	
					
re:Invent 20-Dec		re:Invent 20-Dec		re:Invent 20-Dec	

Case Study

Opening Keynote - Building a smarter and more effective business using AIML on AWS (English Session)

<https://hktw-resources.awscloud.com/ai-machine-learning-web-day/opening-keynote-building-a-smarter-and-more-effective-business-using-aiml-on-aws>

Improving the customer experience of your applications with AWS AI Services (Cantonese session)

<https://hktw-resources.awscloud.com/ai-machine-learning-web-day/improving-the-customer-experience-of-your-applications-with-aws-ai-services>



HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

THANK YOU

